INFRA Watch

Infrastructure Network for Fair Reporting and Accountability

Requirements Analysis Document (RAD)

Infrastructure Network for Fair Reporting and Accountability

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Table of Contents

1. Introduction 7
1.1 Purpose 7
1.2 Scope 7
1.2.1 System Boundary 7
1.2.2 System Context 8
1.2.3 Scope of This Document 8
1.3 Intended Audience 9
1.4 References 9
1.5 Relationship to Other Documents 9
1.6 Document Conventions 10
2. System Overview 11
2.1 Background 11
2.2 System Objectives 11
2.3 System Context 11
2.3.1 High-Level Environment 11
2.3.2 External System Integration 12
2.4 System Architecture Overview 12
2.4.1 Architectural Layers 12
2.4.2 Major Modules and Their Responsibilities 13
2.5 Key System Features 13
2.6 User Roles and Access Context 14
2.7 Operating Environment 15
2.8 Design and Implementation Constraints 15
2.9 Assumptions and Dependencies 16
3. Functional Requirements Analysis 17
3.1 Authentication, Accounts, and Access Control 17
FR-AUTH-01: Registration with Email Verification 17
FR-AUTH-02: Sign-In and Session Management 17
FR-AUTH-03: Password Recovery 18
FR-AUTH-04: Profile and Notification Preferences 18
FR-AUTH-05: Role-Based Access and Moderator Scope 19
FR-AUTH-06: User Administration 19
3.2 Project Data Synchronization 20

FR-SYNC-01: Scheduled ABEMIS Synchronization 20
FR-SYNC-02: Manual Synchronization and History 20
FR-SYNC-03: Scope Filtering of Source Records 21
FR-SYNC-04: Daily Metric Snapshots 21
FR-SYNC-05: Location Reference Sync 22
3.3 Public Transparency Portal 22
FR-PUB-01: Project Directory 22
FR-PUB-02: Project Record 22
FR-PUB-03: Project Map 23
FR-PUB-04: Public Analytics and Regional Statistics 23
FR-PUB-05: Public Content: Articles, Announcements, Notices, Live Broadcasts 24
FR-PUB-06: Information Pages, Language, and Presentation 24
FR-PUB-07: Project Export 25
3.4 Citizen Engagement 25
FR-FB-01: Project Feedback Submission 25
FR-FB-02: Comments, Helpful Votes, Saved Posts 26
FR-FB-03: Feedback Moderation 26
FR-ER-01: E-Report Submission 27
FR-ER-02: E-Report Tracking 27
FR-ER-03: E-Report Handling by Staff 28
FR-UP-01: Evidence Upload and Content Moderation 28
FR-GEO-01: Citizen Evidence Map 29
FR-SMS-01: SMS Grievance Review 29
FR-CON-01: Contact Messages and Suggestions 30
FR-SUR-01: Submission Survey 30
FR-NOT-01: Notifications 31
FR-NOT-02: SLA Follow-Up Reminders 31
3.5 Staff Console and Governance 32
FR-ADM-01: Managerial Dashboard 32
FR-ADM-02: Executive Brief and Managerial Copilot 32
FR-ADM-03: Administrative Project Directory 32
FR-ADM-04: Data Quality Review 33
FR-ADM-05: SLA Reports 33
FR-ADM-06: Audit Log 33
FR-ADM-07: Content and Configuration Management 34

FR-ADM-08: Service Health Check 34
FR-ADM-09: Citizen Engagement Analytics 35
3.6 AI-Assisted Features 35
FR-AI-01: Public Project Assistant (ANIA) 35
FR-AI-02: Knowledge Base 36
FR-AI-03: MYDAS Dashboard Designer 36
FR-AI-04: Voice Assistant 36
3.7 Non-Functional Requirements Analysis 37
NFR-01: Performance and Responsiveness 37
NFR-02: Security and Access Control 37
NFR-03: Reliability and Availability 38
NFR-04: Scalability 38
NFR-05: Maintainability and Quality 39
NFR-06: Usability and Accessibility 39
NFR-07: Interoperability 39
NFR-08: Auditability and Traceability 39
NFR-09: Privacy and Compliance 40
NFR-10: Localization 40
NFR-11: Observability and Operations 40
NFR-12: Deployability and Portability 41
4. Data Requirements Analysis 42
4.1 Purpose 42
4.2 Data Design Overview 42
4.3 Data Entities by Functional Domain 42
4.3.1 Identity and Access 42
4.3.2 Project Mirror, Reference, and Map Data 43
4.3.3 Citizen Engagement 43
4.3.4 Notifications 43
4.3.5 Content and Configuration 44
4.3.6 AI and Knowledge 44
4.3.7 Operations and Platform 44
4.4 Entity Relationships Summary 44
4.5 Data Integrity and Constraints 46
4.6 Data Security Considerations 47
4.7 Data Retention and Archiving 47

4.8 Data Validation Rules 48
5. Interface Requirements Analysis 49
5.1 Purpose 49
5.2 User Interface Requirements 49
5.2.1 Page Inventory 49
5.2.2 Interaction and Design Requirements 49
5.3 External System Interfaces 50
5.3.1 ABEMIS 50
5.3.2 SMS Gateway and SMS Grievance Feed 50
5.3.3 Email, Maps, Storage, Identity, and AI 50
5.4 Hardware Interfaces 51
5.5 Software Interfaces 51
5.6 Communication Protocols and Standards 51
5.7 Interface Security and Compliance 51
5.8 Interface Traceability 51
6. System Constraints and Assumptions 53
6.1 Purpose 53
6.2 Technical Constraints 53
6.3 Environmental Constraints 53
6.4 Operational Constraints 53
6.5 Assumptions 54
6.6 Dependencies 54
6.7 Risks Related to Constraints 54
7. Risk Analysis and Mitigation Plan 55
7.1 Purpose 55
7.2 Risk Assessment Methodology 55
7.3 Risk Categories 55
7.4 Risk Register 55
7.5 Risk Monitoring and Governance 56
7.6 Residual Risk and Contingency 57
8. Requirements Traceability Matrix (RTM) 58
8.1 Purpose 58
8.2 Traceability Methodology 58
8.3 Functional Requirements Traceability 58
8.4 Non-Functional Requirements Traceability 60

8.5 Module, Entity, and Interface Summary 60
9. Appendices 61
9.1 Glossary of Terms 61
9.2 Acronyms 61
9.3 Change History 62
9.4 Document Control 62

1. Introduction

1.1 Purpose

This Requirements Analysis Document (RAD) is the analytical baseline for INFRA Watch, the public transparency and accountability portal for infrastructure projects of the Bureau of Agricultural and Fisheries Engineering (BAFE). It records what the system must do, the data it must hold, the interfaces it must offer, and the constraints it must respect, in a form that developers, testers, and BAFE stakeholders can all work from.

INFRA Watch is built and running. This RAD therefore describes the requirements as they are implemented in the current code base (verified against the repository in October 2026) and marks each requirement with its implementation status. Where a requirement is only partly met, or depends on configuration, the limitation is stated next to the requirement rather than hidden.

The purpose of this document is to:

- Translate the product intent of the portal into verifiable functional and non-functional requirements.
- Give every requirement a stable identifier that can be traced to the routes, server actions, database tables, and acceptance criteria that satisfy it (Chapter 8).
- Define the data that INFRA Watch owns and the data it only mirrors from the ABEMIS source system.
- Document the external systems the portal depends on and the assumptions made about them.
- Record the known risks and open limitations so they can be planned for rather than discovered in production.
- Provide the baseline against which future changes to scope can be assessed through version control.

The companion Software Design Document (SDD) explains how these requirements are realised. Read this RAD for the "what" and the SDD for the "how".

1.2 Scope

1.2.1 System Boundary

INFRA Watch (Infrastructure Network for Fair Reporting and Accountability) lets citizens see where public agricultural and fisheries infrastructure is being built, how far it has progressed, and what it costs, and lets them respond with feedback and reports that BAFE staff can act on. It reads project records from ABEMIS (the Agricultural and Biosystems Engineering Management Information System) for the AMEFIP and INS programs, fiscal years 2021 to 2026.

In scope	Out of scope
Public portal for AMEFIP and INS projects (FY 2021-2026) mirrored from ABEMIS: directory, project record, map, analytics, evidence map.	Creating, validating, approving, or editing projects. ABEMIS remains the system of record; INFRA Watch holds a read model.
Citizen participation: project feedback, comments, helpful votes, E-Reports (issue reports) with geotagged evidence, contact and suggestion forms.	Procurement, contracting, payment, or financial transactions.
SMS grievance review: staff triage of inbound citizen SMS and replies sent by SMS.	Automatic correction of source data. The Data Quality module only reports findings and exports them for follow-up in ABEMIS.

In scope	Out of scope
Staff console: moderation, E-Report handling, scoped analytics, SLA reports, data quality, audit logs, user and content management.	Native mobile applications. The portal is a responsive web application.
AI assistance: public project assistant (ANIA), knowledge base, managerial copilot and executive brief, MYDAS dashboard designer, optional voice assistant.	Visitor identity tracking. Public analytics describe projects; engagement analytics are aggregate and minimal.
Notifications (in-app, email, SMS), scheduled synchronization, background jobs, audit trail.	Ingesting other agencies' projects. The structure allows additional programs, but only BAFE programs are loaded today.

1.2.2 System Context

INFRA Watch sits between the public, BAFE staff, and a small set of external services. Its only upstream data source is the ABEMIS API. Everything else it exchanges with the outside world (SMS, email, maps, storage, AI models) supports citizen participation or staff operations. Figure 1 shows the boundary.

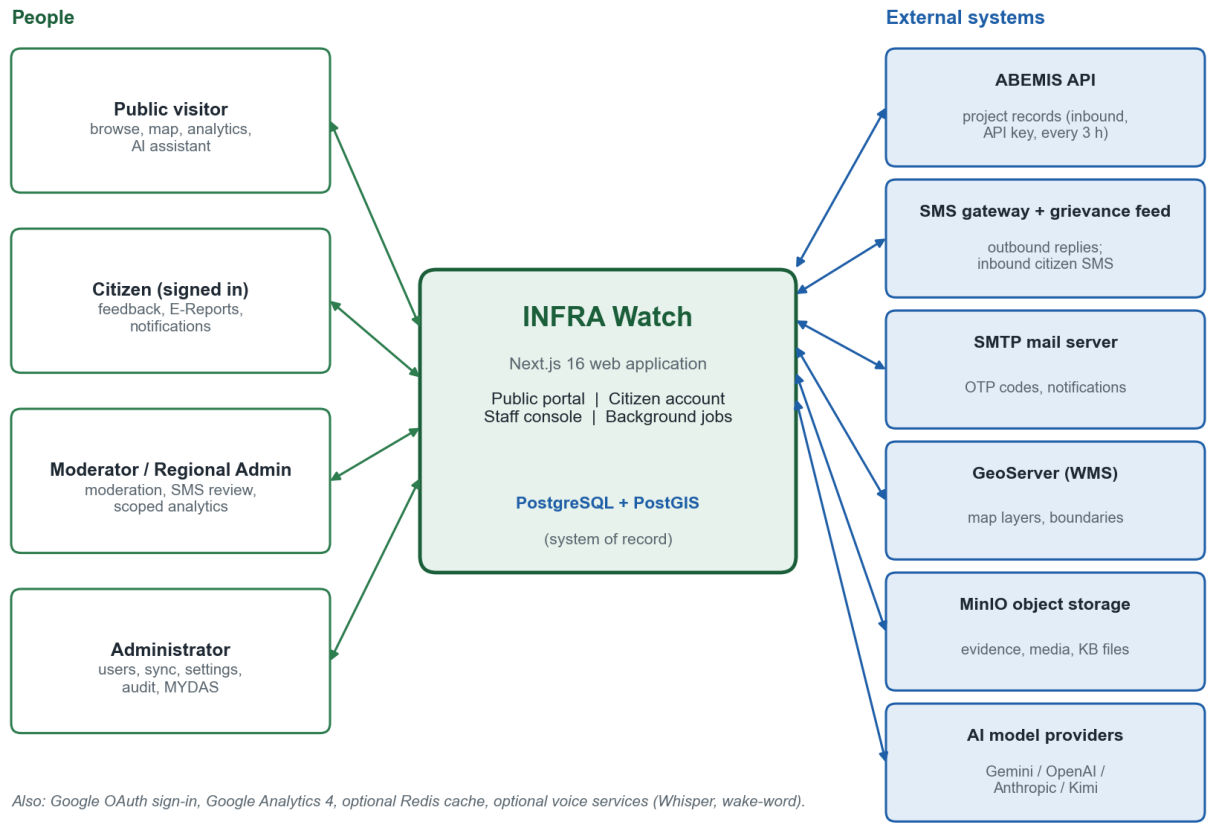


Figure 1. INFRA Watch system context: people on the left, external systems on the right.

1.2.3 Scope of This Document

This RAD covers the functional requirements of every shipped module, the non-functional requirements that constrain them, the data and interface requirements, constraints and assumptions, a risk register, and a requirements traceability matrix. It does not repeat design detail (class structure, algorithms, sequence diagrams); those belong to the SDD.

1.3 Intended Audience

Audience	How they use this document
BAFE program owners and system administrators	Confirm that the portal's behavior matches the mandate; decide priorities for the limitations listed in Chapter 7.
Moderators and regional administrators	Understand what the staff console is expected to let them do, and under what scope restrictions.
Developers (frontend, backend, data)	Use the requirement IDs, data entities, and interface lists as the implementation baseline.
QA and acceptance testers	Derive test cases from the validations, rules, and acceptance criteria in Chapters 3 and 8.
Security and privacy reviewers	Check access control, data handling, and compliance requirements (NFR-02, NFR-09).
Technical maintainers of ABEMIS	See exactly which ABEMIS endpoint and fields INFRA Watch consumes.

1.4 References

Reference	Description
INFRA Watch documentation set (`docs/00` to `docs/13`)	Overview, architecture, database schema, ABEMIS integration, authentication, features, UI/UX contract, API routes, deployment, security, roadmap, migration, mockup plan, AI knowledge retrieval plan.
`docs/current-system/` (feature catalog, workflows, route inventory)	Inventory of implemented functions, state transitions, and routes. Used as a cross-check against the code.
`docs/reviews/code-security-review-2026-09-27.md`	Code, security, and UI review used for the risk register.
ABEMIS 3.0 RAD and SDD	Requirements and design of the upstream source system. INFRA Watch follows their document structure.
Republic Act 10173 (Data Privacy Act of 2012)	Governs collection and handling of personal data from citizens and staff.
8888 Citizens' Complaint Center standard	The 72-hour requirement for an agency to report back on action taken; used as the SLA breach threshold.
WCAG 2.1 Level AA	Accessibility target for public and staff interfaces.
Philippine Standard Geographic Code (PSGC)	Reference data for the region, province, municipality, and barangay hierarchy (`psgc_locations`).
Better Auth, Next.js, Drizzle ORM, PostGIS documentation	Behavior of the framework components the system is built on.

1.5 Relationship to Other Documents

The documents below form a chain. A change to a requirement in this RAD should be reflected in the SDD and, where it affects data or interfaces, in the database schema and route inventory.

Document	Role	Relationship to this RAD
Product overview (`docs/00-overview.md`)	Mission, stakeholders, glossary	Source of the product intent summarized in Chapter 2.
RAD (this document)	What the system must do	Requirement baseline and traceability.
SDD	How the system is built	Design realising each requirement; references the same IDs.
Database schema and ERD (`docs/erd/`)	Authoritative table definitions	Chapter 4 summarizes; the schema files are the source of truth.
Route and API inventory	Endpoint list	Chapter 5 and the RTM cite its routes.
Security review	Findings and patches	Feeds Chapter 7 risks.

1.6 Document Conventions

- **Requirement identifiers** use the form `FR-<MODULE>-NN` for functional requirements and `NFR-NN` for non-functional requirements. Modules: AUTH, SYNC, PUB, FB, ER, UP, SMS, CON, NOT, ADM, AI.
- **Priority** follows MoSCoW: Must, Should, Could. All shipped requirements below are Must or Should.
- **Status** is one of: *Implemented* (works as described), *Partial* (works with a stated limitation), or *Configuration-dependent* (works only when the named environment settings or services are present).
- Identifiers, table names, routes, and environment variables appear in `monospace`.
- "Shall" marks a binding requirement; "may" marks an option.
- Dates and thresholds quoted from code (for example the 72-hour SLA) are the values in force at the document date.

2. System Overview

2.1 Background

BAFE implements agricultural machinery, equipment, facilities, and infrastructure projects under the Agricultural Machinery, Equipment, Facilities, and Infrastructures Program (AMEFIP), including the Irrigation Network Services (INS) sub-program. These projects are recorded in ABEMIS. Until INFRA Watch, a citizen had no simple way to see which projects exist near them, how much was budgeted, how far along they are, or to tell BAFE what they observe on the ground.

INFRA Watch was created to close that gap. It follows the architecture and user-experience patterns of FMR Watch, a sister transparency portal for farm-to-market road projects, but is scoped to AMEFIP and INS data and has since grown its own capabilities: SMS grievance review, an AI project assistant, a knowledge base, managerial analytics, and a data-quality review.

The system is deliberately a **read model plus participation layer**. Project facts come from ABEMIS and are never edited in INFRA Watch. What INFRA Watch owns is citizen input (feedback, comments, votes, reports, evidence), the staff decisions made on that input, and the audit trail of both.

2.2 System Objectives

ID	Objective	Measured by
O1	Make AMEFIP and INS project information (location, budget, status, progress, contractor) publicly visible and searchable.	Directory, map, and analytics pages show current data with a visible last-sync time.
O2	Keep public data current without manual effort.	Scheduled ABEMIS synchronization every three hours; sync history visible to staff.
O3	Let citizens contribute feedback and report problems with verifiable evidence.	Feedback and E-Report submission with geotagged photos and video.
O4	Ensure every citizen submission receives a timely human response.	72-hour SLA thresholds, reminders, and SLA reports.
O5	Give BAFE managers scoped, trustworthy analytics and data-quality visibility.	Managerial dashboard, data-quality findings, explicit unavailable and not-assessed states.
O6	Protect citizens' personal data and keep staff actions accountable.	Role and scope enforcement, anonymity controls, audit log of staff and security events.
O7	Offer assistance that is grounded in the portal's own data.	AI assistant restricted to bounded, read-only tools and approved knowledge.

2.3 System Context

2.3.1 High-Level Environment

INFRA Watch is a single Next.js web application deployed in a Docker container behind an nginx reverse proxy at `infrawatch.bafe.gov.ph`. It stores its data in PostgreSQL with the PostGIS extension and keeps uploaded media in MinIO object storage. Scheduled work (synchronization, reminders, clean-up) runs

inside the same application process and a PostgreSQL-backed job queue. Users reach the system through standard web browsers; no client software is installed.

2.3.2 External System Integration

External system	Direction	Purpose	Mechanism
ABEMIS API	Inbound	Source of all project records	HTTPS GET `/api/infra-amefip-list`, `x-api-key` header, paginated, retried
SMS gateway	Outbound	Replies to citizens; staff and citizen alerts	HTTPS POST with sender ID and credentials from environment
SMS grievance feed	Inbound	Untriaged citizen SMS for staff review	HTTPS GET of a JSON feed (`SMS_GRIEVANCE_API_URL`)
SMTP server	Outbound	OTP codes, notification and reply emails	Nodemailer, STARTTLS on port 587 by default
GeoServer	Outbound (browser)	Map overlays and boundaries	WMS layers configured in `map_layers`
MinIO	Bidirectional	Evidence, media, knowledge-base files	S3-compatible API; public and private buckets
AI model providers	Outbound	Assistant, executive brief, MYDAS	Gemini, OpenAI, Anthropic, or Kimi, selected by `AI_PROVIDER`
Google OAuth	Bidirectional	Optional sign-in	Better Auth social provider, enabled only when credentials exist
Google Analytics 4	Outbound (browser)	Site usage measurement	Enabled only when a measurement ID is built in
Redis	Bidirectional	Optional shared cache for public data	`REDIS_URL`; the system falls back to in-memory cache without it

2.4 System Architecture Overview

2.4.1 Architectural Layers

The system is a layered monolith. Presentation pages call server components and server actions, which call domain services in `lib/`, which use PostgreSQL, MinIO, and the optional Redis cache. The layering is explained in detail in the SDD (Chapter 2); the summary below is what matters for requirements.

Layer	Responsibility	Technology
Presentation	Public, citizen, staff, and authentication pages; maps and charts	Next.js 16 App Router, React 19, Tailwind CSS 4, shadcn/ui, Leaflet, Recharts
Application	Server components, server actions (`actions/query`, `actions/mutation`), route handlers (`app/api`), guards	TypeScript, Zod validation, Better Auth session
Domain services	Synchronization, scope rules, feedback and issue handling, SLA, analytics, AI, notifications, audit	TypeScript modules in `lib/`
Platform and data	Relational and spatial storage, object storage, queue, scheduler, cache	PostgreSQL 18 + PostGIS, Drizzle ORM, MinIO, node-cron, Redis (optional)

2.4.2 Major Modules and Their Responsibilities

Module	Responsibility	Primary code
Authentication and access	Accounts, sessions, roles, permissions, moderator scope	`lib/auth.ts`, `lib/permissions.ts`, `lib/scope.ts`, `lib/session.ts`
Project data sync	Fetch, filter, transform, and store ABEMIS projects; snapshots	`lib/abemis/*`, `lib/scheduler.ts`, `lib/background-worker.ts`
Public portal	Directory, project record, map, analytics, content pages	`app/(public)`, `actions/query/public-projects.query.ts`
Citizen engagement	Feedback, comments, votes, E-Reports, uploads, evidence map	`app/api/projects/[id]/feedback`, `app/api/issues`, `app/api/upload`
SMS grievance	Inbound feed review, tagging, SMS replies	`lib/sms-grievance/*`, `lib/sms.ts`
Staff console	Moderation, issue handling, dashboards, SLA, data quality, audit, user and content admin	`app/(admin)`, `app/api/admin/*`
AI services	Public chat, knowledge base, managerial AI, MYDAS, voice	`lib/chat-*`, `lib/kb-*`, `lib/mydas/*`, `lib/voice`
Notifications	Per-user records, live stream, email and SMS delivery, SLA reminders	`lib/notification-persistence.ts`, `lib/sla-followup.ts`
Audit and monitoring	Audit trail, health check, sync logs	`lib/audit.ts`, `lib/service-health.ts`

2.5 Key System Features

Area	Feature
Transparency	Project directory with search, filters, and CSV/JSON export; project record with budget, status, physical progress, contractor, location, and source media; interactive map with clustering and GeoServer layers; public analytics and regional statistics; citizen evidence map.
Participation	Project feedback with rating, category, optional anonymity, and up to five photos or videos; comments and helpful votes; E-Reports with ticket numbers, geotagged evidence, public tracking, and owner view; contact and suggestion forms; optional submission survey.
Response and accountability	Moderation queues with approve, reject, and delete; public and internal staff responses; 72-hour SLA with reminders; SLA reports; SMS grievance review with real SMS replies; audit log.
Management	Managerial dashboard with scoped filters and drill-through; schedule-health and progress analysis; data-quality findings; executive brief; MYDAS dashboard designer.
Assistance	ANIA public assistant with bounded tools; knowledge base with document upload, FAQ entries, and embeddings; optional admin voice assistant.
Platform	Email OTP verification and recovery, Google sign-in, role and scope enforcement, English and Filipino interface, light and dark theme, background job queue, scheduled synchronization.

2.6 User Roles and Access Context

There are four roles. The first three can use the staff console; they differ in what they may do and how much data they may see. There is no `super_admin` role.

Role	Typical user	Data scope	Console access
Public visitor	Anyone, no sign-in	Public pages and public-safe records only	None
`citizen`	Signed-in resident	Own feedback, issues, notifications, profile	None
`moderator`	Field engineer or local officer	Restricted to the assigned region and/or program; fail-closed if neither is assigned	Yes
`regional_admin`	Regional program manager	Not region-restricted for project data; user management limited by region	Yes
`admin`	BAFE system owner	Unrestricted	Yes, plus admin-only features

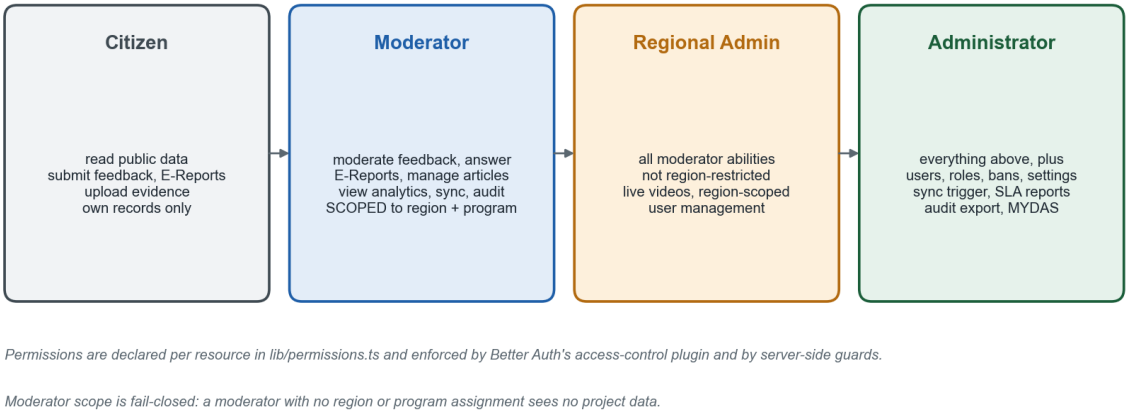


Figure 2. Role capabilities, from least to most privileged.

Permission summary by resource:

Resource	Citizen	Moderator	Regional admin	Admin
Projects	read, list	read, update, list	read, update, list	all, including publish and archive
Feedback	create, read, list	approve, reject, flag, respond, delete	same as moderator	all
Issues (E-Reports)	create, read, list	respond, resolve, update, delete	same as moderator	all
Articles, announcements	read, list	create, publish, unpublish, delete	same as moderator	same
Uploads	create, read	create, read, delete, moderate	same as moderator	same
Contact messages	none	read, update, delete, list	same as moderator	same

Resource	Citizen	Moderator	Regional admin	Admin
Dashboard, analytics	none	view	view	view, export
Reports (SLA)	none	generate, view, export	generate, view, export	adds schedule; SLA pages are admin-only
ABEMIS sync	none	view	view	trigger, view, configure
Data quality, audit logs	none	view	view	audit export and delete as well
Knowledge base	none	all, including embed	all, including embed	all
System settings	none	none	read	read, update
User administration	none	none	list, read, update, ban, revoke (region-scoped)	all, including create, delete, set-role
MYDAS	none	none	none	view, create, export

2.7 Operating Environment

Component	Specification
Application runtime	Bun 1.3 in a Docker container, Next.js 16 standalone output, React 19; runs as a non-root user; ffmpeg included for video checks
Database	PostgreSQL 18 with PostGIS; accessed through the `postgres` driver and Drizzle ORM; pooled connections (default 10)
Object storage	MinIO, one public bucket and one private bucket
Reverse proxy	nginx with Let's Encrypt TLS; request body limit 110 MB; chat endpoint proxied unbuffered
Cache	In-process cache; Redis 7 optional through a Compose profile
Clients	Current evergreen desktop and mobile browsers; the interface is responsive
Time zone	Scheduled jobs run in Asia/Manila

2.8 Design and Implementation Constraints

- **Data source:** project data may only be ingested from the ABEMIS API and only for fiscal years 2021 to 2026. Cancelled, archived, invalid-stage, and pre-implementation records (through 2024) are excluded by rule.
- **Read-only source data:** the system shall not write corrections back to project records or to ABEMIS.
- **Technology:** TypeScript, Next.js App Router, Better Auth, Drizzle ORM, PostgreSQL. Schema changes go through Drizzle migrations.
- **Hosting:** the application container binds to the loopback interface only and is exposed through nginx; PostgreSQL, MinIO, and the reverse proxy are provisioned outside the app's Compose file.
- **Privacy:** citizen personal data is handled under RA 10173; anonymity on public pages shall not remove accountability for staff (the account identifier remains visible to authorized moderators).
- **Language:** user-facing text is available in English and Filipino; plain operational wording is required for data states (unavailable, unknown, stale, cannot be assessed).

- **Interface contract:** UI changes follow `docs/06-ui-ux-design.md` (contrast, 44 by 44 pixel targets, visible focus, responsive tables).

2.9 Assumptions and Dependencies

- ABEMIS exposes `/api/infra-amefip-list` with an API key and stays reasonably available; short outages are absorbed by the three-hourly schedule.
- ABEMIS stage and status values may change spelling; the system maps known variants and falls back to "Not yet started" for unrecognised ones.
- A working SMTP relay is available; without it, verification and recovery codes are not delivered in production.
- Citizens who add photos or video may include GPS metadata; the system reads it in the browser for the evidence map.
- MinIO is reachable. When it is not, uploads fail or fall back to local storage (see risk R-04).
- AI provider keys, embeddings, voice services, and managerial AI are optional and enabled by environment settings.
- BAFE staff assign moderators to a region and program; an unassigned moderator sees no project data by design.

3. Functional Requirements Analysis

Each requirement below states what the system shall do, how the current implementation does it, the data and interfaces involved, and the rules that can be tested. Status and limitations reflect the code base at the document date.

3.1 Authentication, Accounts, and Access Control

Identity is handled by Better Auth with a Drizzle adapter. Roles and permissions are declared once in `lib/permissions.ts` and enforced by the Better Auth admin plugin and by server-side guards.

FR-AUTH-01: Registration with Email Verification

Priority: Must | **Status:** Implemented | **Primary source:** `lib/auth.ts`, `lib/auth-otp-email.ts`

Description: A visitor shall be able to create a citizen account with name, email, and password, and shall verify the email address with a one-time code before the account can be used.

Technical Analysis:

- Better Auth `emailAndPassword` is enabled with `requireEmailVerification: true`; after verification the user is signed in automatically.
- The `emailOTP` plugin issues a 6-digit code that expires after `EMAIL_OTP_EXPIRY` seconds (default 180). The plugin is configured with `disableSignUp: true`, so a code can never create an account by itself.
- In production the code is delivered by SMTP through `deliverAuthCode`; outside production it is printed to the server log only.
- New users receive the role `citizen`. Addresses listed in `ADMIN_EMAIL` or `ADMIN_EMAILS` are created as `admin` (bootstrap administrators).

Behavioral Flow:

1. Visitor submits name, email, password on `/sign-up`.
2. Better Auth creates an unverified `user` and `account` row.
3. A 6-digit code is generated and emailed.
4. Visitor enters the code; `user.email_verified` becomes true.
5. A `session` is created and the user is signed in.

Data Entities: `user`, `account`, `verification`, `session`

Interfaces: `/sign-up`, `/sign-in`, `/api/auth/[...all]`

Validations and Rules:

- Email must be unique and well formed.
- A code is single use and expires; expired codes are rejected.
- Unverified accounts cannot sign in.

FR-AUTH-02: Sign-In and Session Management

Priority: Must | **Status:** Implemented | **Primary source:** `lib/auth.ts`, `lib/session.ts`

Description: A registered user shall be able to sign in with email and password or, when configured, with Google, and shall hold a server-side session until it expires or is revoked.

Technical Analysis:

- Sessions are stored in the `session` table with IP address and user agent, and delivered in a secure cookie handled by the Better Auth Next.js plugin.
- Google OAuth is enabled only when `GOOGLE_CLIENT_ID` and `GOOGLE_CLIENT_SECRET` are present, with the account chooser always shown.
- In production the application refuses to start without `BETTER_AUTH_SECRET` and `BETTER_AUTH_URL`.
- Banned users (`user.banned`) are blocked by the admin plugin; an administrator can terminate all of a user's sessions.
- Failed sign-ins are recorded as audit events.

Data Entities: `session`, `account`, `user`, `audit_logs`

Interfaces: `/sign-in`, `/api/auth/[...all]`

Validations and Rules:

- Only configured origins are trusted for auth requests.
- A banned or deleted user's sessions are rejected on the next request.

FR-AUTH-03: Password Recovery

Priority: Must | **Status:** Implemented | **Primary source:** `app/(auth)/forget-password`, `lib/auth-otp-email.ts`

Description: A user who forgot a password shall be able to reset it using a one-time code sent to the registered email address.

Behavioral Flow:

1. User opens `/forget-password` and requests a code.
2. The code is emailed.
3. User verifies the code.
4. User sets a new password.

Data Entities: `verification`, `account`, `user`

Interfaces: `/forget-password`, `/api/auth/[...all]`

Validations and Rules:

- Codes expire after `EMAIL_OTP_EXPIRY` seconds.
- A password can be reset only with a valid, unexpired code.

FR-AUTH-04: Profile and Notification Preferences

Priority: Must | **Status:** Implemented | **Primary source:** `actions/mutation/profile.mutation.ts`, `lib/account-preferences.ts`

Description: A signed-in user shall be able to view their profile, update their phone number, and choose whether to receive notifications in the app, by email, and by SMS.

Technical Analysis:

- Phone number is an optional user field used for SMS notifications and SMS replies.
- Channel choices are stored per user in `account_preferences` (`in_app`, `email`, `sms`); delivery code consults them before sending.
- Guided tours remember which tours a user has seen in `user_tour_progress`.

Data Entities: `user`, `account_preferences`, `user_tour_progress`

Interfaces: `/my-profile`, `/account-settings`

Validations and Rules:

- Only the owner can change their own profile and preferences.
- Role and scope fields are not user-editable.

FR-AUTH-05: Role-Based Access and Moderator Scope

Priority: Must | **Status:** Implemented | **Primary source:** `lib/permissions.ts`, `lib/scope.ts`, `lib/moderator-scope.ts`, `lib/staff-api.ts`

Description: Every protected operation shall check the caller's role and permission, and a moderator shall only see and act on projects inside their assigned region and program.

Technical Analysis:

- Permissions are declared per resource and action (for example `feedback: approve, reject`) and mapped to four roles: `citizen`, `moderator`, `regional_admin`, `admin`.
- Moderator scope uses two assignments: `user.region` (matched against `projects.region`) and `user.assigned_agency` (matched against `projects.program`, values `AMEFIP` or `INS`).
- Scope is fail-closed: a moderator with neither assignment gets no project data. Regional admins and admins are not restricted by the moderator scope filter.
- Every project query also applies the fiscal-year scope (FR-SYNC-03), so out-of-scope records never surface.
- Staff API routes call `requireStaffApi`; mutating requests also check the request origin.

Data Entities: `user`, `projects`, `feedback`, `issues`

Interfaces: All `/api/admin/*` routes and `(admin)` pages

Validations and Rules:

- Citizens cannot reach staff pages or APIs (HTTP 401/403).
- A scoped moderator cannot read, moderate, or respond to records outside scope, even by guessing an identifier.
- Admin-only features (sync trigger, user administration, settings, SLA reports, MYDAS) are not available to other roles.

FR-AUTH-06: User Administration

Priority: Must | **Status:** Implemented | **Primary source:** `actions/mutation/users.mutation.ts`, `actions/query/users.query.ts`

Description: Administrators shall be able to create, update, ban, unban, verify, and delete users, change roles, assign moderator scope, and terminate sessions, with protection against self-harm.

Technical Analysis:

- Accounts created by an administrator are marked email-verified immediately.
- Role changes, region assignment, program assignment, bans, verification, session termination, and deletion are each audited.
- An administrator cannot change their own role, ban themselves, terminate their own sessions, or delete themselves.
- Regional admins have a narrower user-management capability (list, read, update, ban, revoke) limited by region.

Data Entities: `user`, `session`, `audit_logs`

Interfaces: `/user-management`; Better Auth admin endpoints; `actions/mutation/users.mutation.ts`

Validations and Rules:

- Self-protection checks run on the server, not only in the interface.
- Deleting a user cascades to sessions and accounts.

> **Limitation** Changing a moderator's role does not automatically clear stale region or program assignments; administrators should review scope fields after a role change.

3.2 Project Data Synchronization

Project data is a mirror of ABEMIS. Synchronization keeps the mirror fresh and enforces the scope rules that decide which ABEMIS records belong in INFRA Watch.

FR-SYNC-01: Scheduled ABEMIS Synchronization

Priority: Must | **Status:** Implemented | **Primary source:** `lib/scheduler.ts`, `lib/abemis/client.ts`, `lib/abemis/sync.ts`, `lib/background-worker.ts`

Description: The system shall refresh project data from ABEMIS automatically without staff action.

Technical Analysis:

- A node-cron schedule (default `0 */3 * * *`, Asia/Manila, overridable with `SYNC_CRON_SCHEDULE`) enqueues an `abemis-sync` job in the PostgreSQL job queue.
- A tick is skipped when a sync already succeeded within the last 20 hours, so a healthy day still reaches ABEMIS about once; the frequent schedule exists to recover quickly from upstream outages.
- The worker pages through `/api/infra-amefip-list` 500 records at a time with up to three retries and exponential back-off, then transforms and upserts records in chunks keyed by `abemis_id`.
- A run is reported as completed, completed with warnings (failure rate below 1% of processed records), or failed. A run stuck in `running` for more than 30 minutes is treated as crashed and no longer blocks new runs.

Data Entities: `projects`, `sync_logs`, `background_jobs`

Interfaces: ABEMIS `/api/infra-amefip-list`; scheduler in `lib/scheduler.ts`

Validations and Rules:

- Only one active sync may run at a time.
- The ABEMIS URL must stay on the configured origin and redirects are refused.
- Records written before a failure remain written (incremental design).

FR-SYNC-02: Manual Synchronization and History

Priority: Must | **Status:** Implemented | **Primary source:** `app/api/admin/sync`, `app/(admin)/sync`

Description: Administrators shall be able to start a synchronization on demand, and authorized staff shall be able to see its progress and history.

Technical Analysis:

- `POST /api/admin/sync` requires the `abemis_sync:trigger` permission (admin only) and enqueues the same job as the schedule.
- Moderators and regional admins can view history through `abemis_sync:view`.
- Job progress streams to the browser over server-sent events at `/api/admin/jobs/[id]/stream`.

- Each run records counts of records added, updated, failed, and total processed, plus error details and duration.

Data Entities: `sync_logs`, `background_jobs`

Interfaces: `/sync`, `POST /api/admin/sync`, `GET /api/admin/sync-logs`, `/api/admin/jobs`

Validations and Rules:

- Only admins can trigger a run.
- Duplicate requests are deduplicated by the job's dedupe key.

FR-SYNC-03: Scope Filtering of Source Records

Priority: Must | **Status:** Implemented | **Primary source:** `lib/abemis/year-scope.ts`, `lib/abemis/transform.ts`

Description: Only records inside the INFRA Watch scope shall be stored and shown.

Technical Analysis:

- Fiscal years 2021 to 2026 are in scope (`lib/abemis/year-scope.ts`); the same condition is applied in every project listing and analytics query.
- Exclusion reasons are counted per run: outside-year-scope, proposal-or-pre-implementation-through-2024, cancelled-or-archived, invalid-or-unclassified-stage, and records owned by another portal.
- Raw ABEMIS stages are mapped to three public statuses: Not yet started, On going, Completed.

Data Entities: `projects`

Interfaces: `lib/abemis/year-scope.ts`, `constants/stage-mapping.ts`

Validations and Rules:

- A project with no year funded is out of scope.
- Unmapped stage text falls back to a keyword heuristic, then to Not yet started.

FR-SYNC-04: Daily Metric Snapshots

Priority: Must | **Status:** Implemented | **Primary source:** `lib/analytics/project-metric-snapshots.ts`

Description: After a fully successful sync the system shall record a daily snapshot of each project's progress and budget so trends can be analysed.

Technical Analysis:

- One row per project per capture date holds physical progress, financial progress, budget, ABC, status, and target completion date.
- Snapshots older than 365 days are pruned.
- Snapshots are only captured when the sync completes without failure.

Data Entities: `project_metric_snapshots`, `sync_logs`

Interfaces: Used by `/dashboard` trend and variance views

Validations and Rules:

- Snapshot capture never runs after a failed sync.

FR-SYNC-05: Location Reference Sync

Priority: Should | **Status:** Implemented | **Primary source:** `lib/locations`, `scripts/sync-psgc-locations.ts`

Description: Administrators shall be able to refresh the PSGC location reference data used for region, province, municipality, and barangay filters.

Technical Analysis:

- A `location-sync` job loads `psgc_locations`; the page `/locations/sync` and the script `locations:sync-psgc` start it.
- Location options feed the directory filters, issue forms, and moderator assignment lists.

Data Entities: `psgc_locations`, `background_jobs`

Interfaces: `/locations/sync`, `/api/admin/locations`

Validations and Rules:

- Reference data is replaced as a unit by the job.

3.3 Public Transparency Portal

All public pages are readable without an account. They read the synchronized `projects` table and present explicit unavailable states rather than invented values.

FR-PUB-01: Project Directory

Priority: Must | **Status:** Implemented | **Primary source:** `actions/query/public-projects.query.ts`, `lib/public-project-directory.ts`

Description: Visitors shall be able to search, filter, sort, and page through in-scope projects.

Technical Analysis:

- Search matches project name, ABEMIS identifier, project code, and contractor.
- Filters: program, region, province, municipality, barangay, public status, funding year.
- Sort: latest sync, name, budget, funding year. Pages of 20 records; list or grid view; filter state is serialisable in the URL.
- The page shows the data source and the last successful synchronization time.

Data Entities: `projects`, `sync_logs`

Interfaces: `/projects`, `GET /api/projects`, `GET /api/public/projects`

Validations and Rules:

- Page and limit parameters are bounded; invalid values do not disable pagination.
- Only sanitized public fields are returned.

FR-PUB-02: Project Record

Priority: Must | **Status:** Implemented | **Primary source:** `lib/public-project-record.ts`, `lib/public-source-media.ts`

Description: Visitors shall be able to open a single project and see its details, progress, location, media, and approved feedback.

Technical Analysis:

- A project resolves by ABEMIS identifier, project code, or internal UUID.
- Shown: identity, program, location, approved budget, ABC, stage and public status, dates, reported physical progress, contractor, quantities, source media, coordinates (only when valid), data coverage, and freshness.
- Source media and geotags are sanitized before display; approved feedback and comments appear with the project.

Data Entities: `projects`, `feedback`, `feedback_comments`

Interfaces: `/projects/[id]`, `GET /api/projects/[id]`

Validations and Rules:

- Missing fields display as unavailable, never as zero.
- Coordinates outside the Philippines are not plotted.

> **Limitation** The displayed financial progress value is derived by the source transform and should not be treated as an authoritative disbursement figure until confirmed with ABEMIS.

FR-PUB-03: Project Map

Priority: Must | **Status:** Implemented | **Primary source:** `lib/public-project-map.ts`, `components/map`

Description: Visitors shall be able to explore projects on an interactive map.

Technical Analysis:

- Markers come from the same filters as the directory and are clustered; marker color follows public status.
- Optional GeoServer WMS overlays (boundaries, networks) are listed in `map_layers` and can be toggled; KML sources are fetched through a bounded proxy.
- Selecting a marker opens a bounded project preview with a link to the full record.

Data Entities: `projects` (PostGIS `geom`), `map_layers`

Interfaces: `/map`, `GET /api/map-layers`, `GET /api/kml-proxy`

Validations and Rules:

- Only coordinate pairs inside Philippine bounds are plotted.
- The KML proxy limits file size to 10 MB.

FR-PUB-04: Public Analytics and Regional Statistics

Priority: Must | **Status:** Implemented | **Primary source:** `actions/query/analytics.query.ts`, `lib/public-analytics`

Description: Visitors shall be able to see portfolio-level analytics for AMEFIP and INS projects.

Technical Analysis:

- `/infra-analytics` reports project counts by stage, approved budget total and coverage, completion and turnover summary, mapped-project coverage, region totals, and banner-program totals, with source and freshness shown.
- `/statistics` shows regional statistics (funded, completed, ongoing, not yet started, suspended, allocated amount) by year and program from `regional_statistics`.

- Aggregation refuses to run beyond 30,000 rows and returns an explicit unavailable state.

Data Entities: `projects`, `regional_statistics`

Interfaces: `/infra-analytics`, `/statistics`, `GET /api/public/regional-statistics`

Validations and Rules:

- Results are cached briefly (regional statistics: 60 seconds).
- These are project analytics, not visitor analytics.

FR-PUB-05: Public Content: Articles, Announcements, Notices, Live Broadcasts

Priority: Must | **Status:** Implemented | **Primary source:** `actions/query/live-videos.query.ts`, `lib/live-video-*`

Description: Visitors shall see current articles, announcements, site notices, and live or recorded broadcasts managed by staff.

Technical Analysis:

- Articles can link to a project; announcements and site notices appear only between their start and end dates.
- Live entries may be Facebook Live, YouTube, or uploaded recordings; at most one live and one featured entry is allowed at a time.
- Inactive, unpublished, or expired entries are hidden from public queries.
- A site-wide maintenance notice and optional banner (from `system_settings`) are served by `/api/public/site-notices`.

Data Entities: `articles`, `announcements`, `live_videos`, `system_settings`

Interfaces: `/articles-and-updates`, `/live`, `GET /api/public/articles`, `GET /api/announcements/active`, `GET /api/public/site-notices`

Validations and Rules:

- Provider URLs must be HTTPS.
- Schedule and state combinations are validated before saving.

FR-PUB-06: Information Pages, Language, and Presentation

Priority: Should | **Status:** Implemented | **Primary source:** `i18n/`, `components/cookie-consent`

Description: The portal shall provide About, FAQ, Contact, Data Privacy, Data Deletion, Terms of Service, and SMS guide pages in English and Filipino, with a light and dark theme and a cookie-consent notice.

Technical Analysis:

- The interface language is switchable at any time and remembered in the browser.
- Theme choice is stored in browser local storage; the layout is responsive for phone, tablet, and desktop.
- Essential cookies are always on; analytics cookies stay off until the visitor opts in through the consent banner.

Data Entities: None (static and translation files)

Interfaces: `/about`, `/faq`, `/contact`, `/data-privacy`, `/data-deletion`, `/terms-of-service`, `/citizen-guides/sms`

Validations and Rules:

- All user-facing strings exist in both languages.
- Pages remain usable when JavaScript storage is unavailable.

FR-PUB-07: Project Export

Priority: Should | **Status:** Implemented | **Primary source:** `lib/project-export.ts`

Description: Visitors shall be able to download the filtered project list.

Technical Analysis:

- `GET /api/public/project-export` returns JSON, or CSV (`format=csv`) as `infrawatch-projects.csv`.
- The export applies the same fiscal-year scope and filters as the directory.

Data Entities: `projects`

Interfaces: `GET /api/public/project-export`

Validations and Rules:

- Exports contain public fields only.

3.4 Citizen Engagement

Citizen participation has two channels: project **feedback** (opinions and observations tied to a project, shown publicly after approval) and **E-Reports** (issues that need action, tracked by ticket). A third channel, SMS, reaches citizens without internet access.

FR-FB-01: Project Feedback Submission

Priority: Must | **Status:** Implemented | **Primary source:** `app/api/projects/[id]/feedback/route.ts`, `actions/query/settings.query.ts`

Description: A signed-in citizen shall be able to submit feedback about a specific project, optionally with a rating, photos, or video, and optionally shown anonymously.

Technical Analysis:

- The citizen selects an existing project, chooses a category (general, quality, progress, concerns), writes a comment, optionally gives a 1 to 5 rating, optionally attaches up to five images or videos, and may request public anonymity. The server accepts comments of up to 1,000 characters.
- The server validates session, project, content (text is screened for unacceptable language), media paths, and category, then stores the record and audits the event.
- The initial status follows the `auto_reply_config` system setting. With auto-accept on (the default), the feedback is stored as `approved` and the citizen receives an automated acknowledgment at once, so moderation happens after publication. With it off, the feedback is stored as `pending` until a moderator decides. The earlier five-minute delayed acknowledgment job has been retired.
- Scoped staff are notified. Only approved feedback is shown publicly; the citizen can follow status and moderation notes in `/my-feedbacks`.

Behavioral Flow:

1. Citizen opens `/citizen-feed` or a project page and starts feedback.
2. Media is uploaded through the validated upload endpoint (FR-UP-01).
3. The feedback is created as `approved` (auto-accept on) or `pending` (off), with the user's identifier.
4. Staff are notified and the citizen is acknowledged.

5. A moderator can approve, reject, or delete it at any time (FR-FB-03).

Data Entities: `feedback`, `projects`, `notifications`, `audit_logs`

Interfaces: `POST /api/projects/[id]/feedback`, `/citizen-feed`, `/my-feedbacks`, `GET /api/my-feedbacks`

Validations and Rules:

- Anonymity hides the author publicly only; the user identifier stays stored and visible to authorized moderators.
- Media must come from the portal's own upload paths.
- Owners may edit or delete their own feedback.

> **Limitation** The feedback composer allows up to 2,000 characters but the server rejects comments over 1,000, so the limits should be aligned. With auto-accept on, text and media are published after automated checks only; moderators review afterwards.

FR-FB-02: Comments, Helpful Votes, Saved Posts

Priority: Should | **Status:** Implemented | **Primary source:** `lib/comment-moderation.ts`

Description: Signed-in users shall be able to comment on approved feedback, mark comments helpful or unhelpful, and save or subscribe to posts.

Technical Analysis:

- Comments may include portal-hosted media and are checked for profanity; flagged comments are auto-rejected with a reason and enter a quarantine list for manual review.
- Votes are stored per user and comment (`vote_type`); counts are kept on the comment.
- `saved_posts` and `post_subscriptions` support bookmarks and follow-up notifications.

Data Entities: `feedback_comments`, `feedback_comment_votes`, `saved_posts`, `post_subscriptions`

Interfaces: Server actions in `actions/mutation/feedback-comment.mutation.ts`, `feedback-vote.mutation.ts`, `post-interactions.mutation.ts`

Validations and Rules:

- A user has at most one vote per comment.
- Comment length is capped at 10,000 characters.

FR-FB-03: Feedback Moderation

Priority: Must | **Status:** Implemented | **Primary source:** `app/api/admin/feedback`, `app/(admin)/feedbacks`

Description: Moderators and administrators shall be able to review, approve, reject, and delete feedback and comments within their scope.

Technical Analysis:

- `/feedbacks` lists scoped feedback with statistics and search by comment, submitter, project, or identifier.
- Approve or reject records the moderator, timestamp, and an optional note; rejection reasons are visible to the owner.
- Staff may permanently delete feedback and its attachments; deletion is audited.
- Moderation of an already decided record is permitted, so staff can reject or delete feedback that was auto-approved.

- Comments are moderated separately in `/comment-moderation`.

Data Entities: `feedback`, `feedback_comments`, `audit_logs`, `notifications`

Interfaces: `/feedbacks`, `/comment-moderation`, `/api/admin/feedback`, `/api/admin/feedback/[feedbackId]/moderate`, `/api/admin/comments`

Validations and Rules:

- Permission and scope are checked on every request.
- Every decision writes an audit record and updates public and citizen views.

FR-ER-01: E-Report Submission

Priority: Must | **Status:** Implemented | **Primary source:** `app/api/issues/route.ts`, `lib/report-issue-validation.ts`

Description: Any visitor shall be able to submit an issue report about infrastructure, anonymously or with contact details, with optional project link, location, and evidence.

Technical Analysis:

- The reporter chooses anonymous or identified mode; identified reports require contact information. A signed-in reporter's user identifier is stored.
- An issue type from the shared catalog (for example quality, progress, safety, equipment) and a description are required; location may be region, province, municipality, barangay, landmark, and coordinates.
- Evidence may be images or video, including geo-video with a GPS track sidecar.
- A unique ticket number is generated and scoped staff are notified; the event is audited.
- The initial status follows the `auto_reply_config` setting. With auto-accept on (the default) the report starts as `reviewing`, the description is published as the public description when it has at least 20 characters, and an automated response from "InfraWatch Automated System" is recorded. With it off the report starts as `pending` and waits for staff.

Data Entities: `issues`, `projects`, `notifications`, `audit_logs`

Interfaces: `/report-issue/new`, `POST /api/issues`

Validations and Rules:

- Coordinates must be valid and inside the Philippines to be used.
- The project link is optional.
- Upload and evidence rules in FR-UP-01 apply.

FR-ER-02: E-Report Tracking

Priority: Must | **Status:** Implemented | **Primary source:** `lib/public-issue-dto.ts`, `lib/public-issue-activity.ts`

Description: Anyone shall be able to see public-safe E-Report progress, and a signed-in reporter shall see their own reports in more detail.

Technical Analysis:

- Public lists and details expose only a public description, public-safe evidence and location, activity, and public staff responses; contact details, internal notes, and unapproved text are removed by a dedicated public DTO.
- `/my-issues` shows the owner's reports with status counts, evidence, responses, and history.

Data Entities: `issues`, `issue_responses`

Interfaces: `/report-issue`, `/report-issue/[id]`, `/my-issues`, `GET /api/issues`, `GET /api/my-issues`

Validations and Rules:

- Internal-only responses never appear in public or owner timelines.
- Owner access requires a matching session.

FR-ER-03: E-Report Handling by Staff

Priority: Must | **Status:** Implemented | **Primary source:** `app/api/admin/issues`, `lib/admin-issues.ts`

Description: Moderators and administrators shall be able to respond to, update, publish, and resolve E-Reports within their scope.

Technical Analysis:

- `/issues` lists scoped reports with status totals, search, and pagination; the detail view shows private reporter information and all evidence.
- A response may be public or internal-only and may change status among `pending`, `reviewing`, `resolved`, `closed`; moving to `resolved` sets `resolved_at`.
- A public description can be approved for publication separately (`public_approved_at`, `public_approved_by`).
- Staff can approve or withdraw each evidence image for public display; an image is shown only after the report has a published public summary.
- Staff can permanently delete a report, its evidence references, and responses.

Data Entities: `issues`, `issue_responses`, `audit_logs`

Interfaces: `/issues`, `/issues/[id]`, `/api/admin/issues`, `POST /api/admin/issues/[id]/responses`

Validations and Rules:

- Every action is audited and the reporter is notified of public responses.
- Evidence reads for staff outside scope are denied.

> **Limitation** The server accepts any recognised target status from any other status; there is no enforced transition graph, and response insertion, status update, audit, and notification are not committed in a single transaction.

FR-UP-01: Evidence Upload and Content Moderation

Priority: Must | **Status:** Implemented | **Primary source:** `lib/upload-policy.ts`, `lib/upload-validation.ts`, `lib/video-moderation.ts`

Description: The system shall accept images and video as evidence, validate them strictly, check them for inappropriate content, and store them safely.

Technical Analysis:

- Allowed types: JPEG, PNG, WebP, GIF images; MP4, MOV, WebM video. Size limits are set by `MAX_IMAGE_SIZE` and `MAX_VIDEO_SIZE`; large video uses resumable multipart upload with 50 MB parts.
- The server requires a session, an allowlisted destination folder, and a matching MIME type, extension, and file signature.
- Optional NSFW image and sampled-video-frame moderation runs with configurable thresholds and a fail-closed option.

- Files are stored in MinIO; public media prefixes are readable publicly, while issue evidence and knowledge-base files use a private bucket served through authenticated handlers.
- Blocked uploads are written to the audit log.

Data Entities: `upload_sessions`, `audit_logs`

Interfaces: `POST /api/upload`, `/api/upload/multipart`, `/api/upload/preview`

Validations and Rules:

- Rejected files are never stored.
- Evidence for an E-Report is visible only to its owner, administrators, and in-scope moderators.

> **Limitation** Stored images are not re-encoded, so EXIF metadata (including GPS) remains in the file. The code can fall back to local storage if MinIO is unreachable, which is not durable in a container.

FR-GEO-01: Citizen Evidence Map

Priority: Should | **Status:** Implemented | **Primary source:** `components/shared/system-evidence-map-client.tsx`

Description: Visitors shall be able to see geolocated public evidence from approved feedback and public E-Reports on a map.

Technical Analysis:

- `GET /api/evidence` returns public-safe records with sanitized media and coordinates.
- The browser can extract GPS coordinates from photos to propose a location to the citizen.

Data Entities: `feedback`, `issues`

Interfaces: `/evidence-map`, `GET /api/evidence`

Validations and Rules:

- Only approved or public-safe records appear.

FR-SMS-01: SMS Grievance Review

Priority: Must | **Status:** Implemented | **Primary source:** `lib/sms-grievance/*`, `lib/sms.ts`

Description: Staff shall be able to review citizen SMS grievances, tag them to a BAFE project or mark them out of scope, and reply by SMS.

Technical Analysis:

- Untriaged messages are loaded server-side from the SMS grievance feed (`SMS_GRIEVANCE_API_URL`); a clearly labelled synthetic fallback is used only if the feed cannot be reached.
- The original text is never altered. Staff decisions (project tag, category, routing, status, notes, SMS thread) are stored in `sms_grievance_reviews` with optimistic version checks so two reviewers cannot send the same reply.
- Tagging a project, marking Not a BAFE project, linking a follow-up, and the Reply tab each send a real SMS through `lib/sms.ts`; the decision is saved first and a failed send is shown as failed. Internal notes are never sent.
- An optional first-reply job (`SMS_AUTO_ACK_ENABLED`, `SMS_AUTO_ACK_SINCE`) asks each new sender for project type, name, age, gender, and location, at most once per message.

Data Entities: `sms_grievance_reviews`

Interfaces: `/issues/sms-review`, `/issues/sms-review/[id]`, `/api/admin/sms-grievance/attention`, `actions/mutation/sms-grievance.mutation.ts`

Validations and Rules:

- A parsed project phrase is only a hint; staff must pick an actual project record.
- Senders without a mobile number can be reviewed but not texted.
- Sample fixtures never contain real phone numbers.
- A per-minute job folds a sender's new message into their existing case (within a week) and never sends an SMS.

> **Limitation** SMS reviews do not yet create rows in `issues`; case tracking happens on the review pages only.

FR-CON-01: Contact Messages and Suggestions

Priority: Should | **Status:** Implemented | **Primary source:** `lib/contact-messages.ts`, `lib/suggestions.ts`

Description: Visitors shall be able to send a message or suggestion to BAFE, and staff shall be able to read and answer it.

Technical Analysis:

- Suggestion categories: suggestion, website problem, general comment; a hidden honeypot field discourages bots.
- Staff read messages in `/contact-messages` and send replies recorded in `contact_responses` (status and sent time).
- Each message stores a sender key and a handling status.

Data Entities: `contact_messages`, `contact_responses`

Interfaces: `/contact`, `/suggestions`, `/contact-messages`, `/reports/contact`

Validations and Rules:

- Subject and message have length limits (subject 160, message 5,000 characters).
- Email addresses are validated.

FR-SUR-01: Submission Survey

Priority: Could | **Status:** Implemented | **Primary source:** `hooks/use-submission-survey-gate.ts`

Description: After a first submission the system may ask the citizen a short, skippable survey to understand who uses the portal.

Technical Analysis:

- Answers (respondent type, age, gender, referral source) are stored with the source submission and user in `submission_surveys`; a `skipped` flag records a decline.
- The gate prevents repeated prompting.

Data Entities: `submission_surveys`

Interfaces: `/api/submission-survey`, `/api/submission-survey/status`

Validations and Rules:

- The survey is optional and never blocks submission.
- Demographic answers are aggregated for reporting.

FR-NOT-01: Notifications

Priority: Must | **Status:** Implemented | **Primary source:** `lib/notification-persistence.ts`, `lib/realtime-notifications.ts`

Description: Users shall receive notifications about events that concern them through the channels they choose.

Technical Analysis:

- A domain event creates one `notifications` record and per-user `notification_recipients` rows in a transaction; recipients are chosen by project scope and the user's channel preferences.
- In-app delivery is live through authenticated server-sent events at `/api/notifications/stream`; email and SMS delivery run as `notification-delivery` jobs.
- Citizens see history at `/my-notifications` and can mark items read.

Data Entities: `notifications`, `notification_recipients`, `notification_reads`, `account_preferences`, `background_jobs`

Interfaces: `/my-notifications`, `GET /api/notifications`, `GET /api/notifications/stream`

Validations and Rules:

- Orphaned notifications are cleaned when a user is removed.
- Failed external delivery is retried by the job queue.

> **Limitation** Live fan-out uses an in-process connection registry, so with more than one application replica a connected user only receives live events produced by the replica they are connected to; stored history is always complete.

FR-NOT-02: SLA Follow-Up Reminders

Priority: Must | **Status:** Implemented | **Primary source:** `lib/sla-thresholds.ts`, `lib/sla-followup.ts`

Description: Staff shall be reminded before a citizen submission breaches the 72-hour response standard.

Technical Analysis:

- The threshold follows the 8888 Citizens' Complaint Center standard: 72 hours. A gentle reminder goes to scoped moderators at 24 hours, an urgent reminder to moderators and admins at 60 hours, and a breach notice to admins after 72 hours.
- The same constants drive the SLA reports so both always agree.
- A job runs every 15 minutes and stamps each checkpoint on the record to avoid repeats. With auto-accept on, few items wait as pending, so reminders mostly concern issues awaiting a first public response.

Data Entities: `issues`, `feedback` (checkpoint timestamps), `notifications`

Interfaces: `lib/sla-followup.ts` (scheduled)

Validations and Rules:

- Feedback counts as answered on a real moderation decision, not on automatic acknowledgment.
- Issues count as answered on the first non-internal response.

> **Limitation** A checkpoint timestamp may be written before external delivery succeeds, which can prevent an automatic retry after a delivery failure.

3.5 Staff Console and Governance

FR-ADM-01: Managerial Dashboard

Priority: Must | **Status:** Implemented | **Primary source:** `lib/analytics/managerial-dashboard-query.ts`, `lib/analytics/schedule-health.ts`

Description: Authorized staff shall see portfolio KPIs and charts for the projects in their scope, and be able to drill from a chart to the projects behind it.

Technical Analysis:

- Filters are parsed with a strict schema; invalid filters are rejected. Scope (moderator region and program) and row limits are enforced before aggregation.
- Views: totals and budget coverage, schedule health calculated from dates and progress, progress and delayed projects, priority projects, region performance, project-type budgets, funding years, trend snapshots, and progress variance.
- Unknown, stale, and cannot-be-assessed states are shown explicitly.
- Chart selections update URL filters or run a bounded drill-through query.

Data Entities: `projects`, `project_metric_snapshots`

Interfaces: `/dashboard`, `/api/admin/analytics`, `/breakdown`, `/drillthrough`, `/progress-curve`, `/cost-by-region`

Validations and Rules:

- Moderators see only in-scope data.
- Row limits protect the database from unbounded aggregation.

FR-ADM-02: Executive Brief and Managerial Copilot

Priority: Could | **Status:** Configuration-dependent | **Primary source:** `lib/analytics/executive-brief.ts`, `lib/analytics/managerial-ai-*.ts`

Description: When enabled, authorized staff shall be able to ask questions of their dashboard data and generate a management brief.

Technical Analysis:

- Enabled only when `ENABLE_MANAGERIAL_AI=true`; uses the same authorization scope as the dashboard.
- Generated text keeps a verification notice and the authorized-data scope notice.
- Conversation history is stored server-side with role-based visibility.

Data Entities: `ai_chat_history`

Interfaces: `/executive-brief`, `POST /api/admin/analytics/assistant`

Validations and Rules:

- The model receives bounded context and tools only.
- Output is advisory and must be verified.

FR-ADM-03: Administrative Project Directory

Priority: Should | **Status:** Implemented | **Primary source:** `app/api/admin/projects`

Description: Staff shall be able to search and filter the synchronized project records in their scope.

Technical Analysis:

- Same scope rules as the dashboard; shows portfolio statistics and paginated records.
- The page is a read model view, not a project authoring tool.

Data Entities: `projects`

Interfaces: `/admin-projects`, `GET /api/admin/projects`, `/stats`

Validations and Rules:

- Scope applies to every query.

FR-ADM-04: Data Quality Review

Priority: Must | **Status:** Implemented | **Primary source:** `lib/data-quality/*`

Description: Staff shall be able to see where source project data is missing, inconsistent, stale, or implausible, and export the findings for follow-up in ABEMIS.

Technical Analysis:

- Findings are computed from synchronized records and filtered, searched, and paginated in the page.
- Export produces a list for source-system owners. The module never edits project data; the project `PATCH` endpoint deliberately returns HTTP 405.
- Manual corrections, if any, are recorded separately in `project_data_corrections` and `project_commodity_overrides`.

Data Entities: `projects`, `project_data_corrections`, `project_commodity_overrides`

Interfaces: `/data-quality`, `/api/admin/data-quality`, `/export`

Validations and Rules:

- Recommendation-only: no automatic repair.

FR-ADM-05: SLA Reports

Priority: Must | **Status:** Implemented | **Primary source:** `actions/query/reports.query.ts`

Description: Administrators shall be able to measure how quickly issues and feedback are answered.

Technical Analysis:

- For a chosen date range the report shows average, median, minimum, and maximum response time, distribution, daily trend, resolution rate, and breach flags.
- Reports can be downloaded from the page.

Data Entities: `issues`, `issue_responses`, `feedback`

Interfaces: `/reports/issues`, `/reports/feedbacks`

Validations and Rules:

- Admin only.
- Automatic acknowledgment is excluded from feedback response time.

FR-ADM-06: Audit Log

Priority: Must | **Status:** Implemented | **Primary source:** `lib/audit.ts`

Description: Authorized staff shall be able to see who did what and when, including security events.

Technical Analysis:

- Records table, record identifier, action, actor, old and new values, changed fields, IP address, user agent, and notes.
- Covers record changes, failed logins, blocked and NSFW-rejected uploads, and sync activity.
- The page filters by group, action, category, date, and search text.

Data Entities: `audit_logs`

Interfaces: `/audit-logs`, `GET /api/audit-logs`

Validations and Rules:

- View is limited to authorized staff; export and delete are admin-only.
- Writing an audit record never blocks the user action it describes.

FR-ADM-07: Content and Configuration Management

Priority: Should | **Status:** Implemented | **Primary source:** `actions/mutation/content.mutation.ts`

Description: Staff shall be able to manage articles, announcements, email templates, map layers, live videos, regional statistics, and system settings.

Technical Analysis:

- Articles and announcements are available to moderators and above; live videos to regional admins and admins; system settings are read by regional admins and changed by admins.
- `/settings` holds the auto-accept (auto-reply) configuration for feedback, issues, and contact messages, a maintenance-mode switch, and a public banner. During maintenance the public sees a notice while staff keep working and the sign-in pages stay reachable; the banner is checked for AA text contrast.
- Email templates are stored with variables and an active flag; map layers store the GeoServer workspace, layer, style, opacity, legend, and order.

Data Entities: `articles`, `announcements`, `email_templates`, `map_layers`, `live_videos`, `regional_statistics`, `system_settings`

Interfaces: `/articles`, `/announcements`, `/email-templates`, `/map-layers`, `/live-videos`, `/regional-statistics`, `/settings`

Validations and Rules:

- Every change is audited.
- Only HTTPS provider URLs are accepted for live videos.

FR-ADM-08: Service Health Check

Priority: Should | **Status:** Implemented | **Primary source:** `lib/service-health.ts`

Description: Administrators shall be able to view the health of dependent services and clear the public cache.

Technical Analysis:

- `GET /api/health` returns service health to staff; `DELETE /api/health` clears this instance's public cache, checks the request origin, and writes an audit record.

Data Entities: `audit_logs`

Interfaces: `/health-check`, `/api/health`

Validations and Rules:

- Staff only; mutation checks origin.

FR-ADM-09: Citizen Engagement Analytics

Priority: Could | **Status:** Implemented | **Primary source:** `lib/analytics/citizen-event-*.ts`

Description: Staff shall be able to see aggregate, privacy-minimal usage of the portal.

Technical Analysis:

- Events record an event name, time, route template, resource type and identifier, entry surface, a banded result count, and a coarse network region. No visitor identity is stored.
- Daily aggregates feed the report at `/reports/citizen-engagement`; maintenance prunes old events.
- Per-client rate limiting protects the event endpoint.

Data Entities: `analytics_events`, `analytics_daily_aggregates`

Interfaces: `POST /api/analytics/events`, `/reports/citizen-engagement`

Validations and Rules:

- Events are bounded in size and rate.
- Reports show aggregates only.

3.6 AI-Assisted Features

AI features are assistants, not authorities. They answer from the portal's own data through bounded tools, refuse out-of-scope requests, and are limited by rate and request-size controls.

FR-AI-01: Public Project Assistant (ANIA)

Priority: Must | **Status:** Configuration-dependent | **Primary source:** `app/api/chat/route.ts`, `lib/chat-*.ts`, `lib/ai-provider.ts`

Description: Visitors shall be able to ask questions about public infrastructure projects and receive grounded answers with suggested next steps.

Technical Analysis:

- `POST /api/chat` accepts a bounded request (64 KB, 20 history messages, 4,000 characters per message) and streams the answer.
- The model can call only five read-only tools: `searchProjects`, `getProjectStats`, `getProjectById`, `getDelayedProjectsSummary`, `searchKnowledgeBase`.
- A policy layer refuses unrelated requests, attempts to reveal internal details or user accounts, and prompt-injection phrases; retrieved data is treated as untrusted text.
- Per-minute and global daily limits are stored in `ai_chat_rate_limits`; `CHAT_REQUIRE_AUTH` can force sign-in.
- History is stored server-side and purged daily after the retention period.

Data Entities: `ai_chat_history`, `ai_chat_rate_limits`, `projects`, `kb_chunks`

Interfaces: `POST /api/chat`

Validations and Rules:

- The model never receives raw user identity.
- Answers only concern public AMEFIP and INS project information.

FR-AI-02: Knowledge Base

Priority: Must | **Status:** Implemented | **Primary source:** `lib/kb-*.ts`

Description: Staff shall be able to maintain the documents and FAQ entries the assistant can draw on.

Technical Analysis:

- Staff upload supported documents or enter FAQ question and answer pairs; text is extracted, chunked, embedded locally (384 dimensions), and stored as JSON in `kb_chunks`.
- Documents can be archived, restored, reindexed, downloaded, and permanently deleted after archiving. Archived documents are never retrieved.
- Reindexing keeps the old chunks until replacements are ready, then swaps them transactionally.
- Retrieval ranks chunks by cosine similarity computed in the application.

Data Entities: `kb_documents`, `kb_chunks`

Interfaces: `/knowledge-base`, `POST /api/knowledge-base/upload`, `/process`, `GET /api/knowledge-base/files/[id]`

Validations and Rules:

- Processing requires the `knowledge_base:embed` permission.
- Files live in the private bucket and are read through authenticated handlers.

> **Limitation** There is no separate draft or published state: every active, embedded document is eligible for public assistant retrieval, so confidential material should not be uploaded.

FR-AI-03: MYDAS Dashboard Designer

Priority: Could | **Status:** Configuration-dependent | **Primary source:** `lib/mydas/*`

Description: Administrators shall be able to describe an analysis in plain language and receive a chart built from a safe, read-only query.

Technical Analysis:

- The model proposes a SQL query and display type from a catalog of chart types; every query passes a validator before execution.
- The validator allows a single `SELECT`, rejects forbidden keywords and multiple statements, restricts access to an allowlist of tables, and enforces a row limit (default 500, maximum 1,000).
- Dashboards are saved in `mydas_dashboards` and can be exported.

Data Entities: `mydas_dashboards`

Interfaces: `/mydas`, `actions/mutation/mydas.mutation.ts`

Validations and Rules:

- Admin only.
- Generated SQL is never trusted: validation is the security boundary.

FR-AI-04: Voice Assistant

Priority: Could | **Status:** Configuration-dependent | **Primary source:** `lib/voice`, `hooks/use-voice-assistant.ts`

Description: When enabled, administrators shall be able to talk to the assistant using a wake word.

Technical Analysis:

- Controlled by `VOICE_ASSISTANT_ENABLED`; issues signed wake-word tokens, sends bounded audio to a private Whisper-compatible endpoint, and speaks answers using an in-browser speech model with a browser fallback.

Data Entities: None

Interfaces: `POST /api/voice/wake-token`, `POST /api/voice/transcribe`

Validations and Rules:

- Admin only.
- Audio size is bounded.

> **Limitation** The wake-word and Whisper services are not defined in the production Compose file and must be provisioned separately.

3.7 Non-Functional Requirements Analysis

NFR-01: Performance and Responsiveness

Description: Public pages and APIs shall remain responsive under normal load.

Technical Analysis:

- List endpoints are paginated with bounded page sizes (directory 20 per page; admin lists 10 to 100).
- Aggregations refuse to run beyond 30,000 rows and return an explicit unavailable state.
- Public reads are cached (in memory or Redis) for short periods; regional statistics for 60 seconds; cache is namespaced per instance and can be cleared by an admin.
- Reference indexes exist on project identifier, province, region, status, and funding year; issues and feedback are indexed by status and project.
- Large video upload is chunked into 50 MB parts.

Validation Metrics:

Metric	Target	Measured by
Directory and record pages	usable within a few seconds on a typical connection	browser performance tools
Aggregation queries	bounded by row limit	query tests
Chat request	64 KB maximum; response timeout `CHAT_RESPONSE_TIMEOUT_MS`	unit tests

NFR-02: Security and Access Control

Description: The system shall protect data and functions from unauthorised access and common web attacks.

Technical Analysis:

- Role and permission checks on every protected route and server action; moderator scope fail-closed (FR-AUTH-05).
- Secure session cookies, required secrets in production, trusted-origin list, origin check on staff mutations.

- Strict upload validation, content moderation, and private storage for evidence and knowledge-base files.
- AI safety: bounded tools, refusal policy, SQL allowlist and validator for MYDAS.
- ABEMIS client refuses redirects and cross-origin endpoints; KML proxy is size-limited.
- Proxy headers: nginx replaces client-supplied forwarding headers; rate limiting uses a trusted-proxy setting.

Validation Metrics:

Metric	Target	Measured by
Unauthenticated staff request	401 or 403	route tests
Out-of-scope record access	denied	scope tests
Invalid upload	rejected and audited	upload tests

NFR-03: Reliability and Availability

Description: Temporary failures of dependencies shall not take the portal down or corrupt data.

Technical Analysis:

- Cron-scheduled sync every three hours tolerates ABEMIS outages; failed runs do not block later runs after a 30-minute stale window.
- The job queue uses leases, heartbeats, retry with back-off, and dedupe keys, so a crashed worker's job is reclaimed.
- The Docker service restarts unless stopped; start-up waits for the database and applies migrations.
- Redis is optional: without it the cache falls back to memory.

Validation Metrics:

Metric	Target	Measured by
Sync run reliability	tolerates up to 1% record failures as warnings	sync logs
Job recovery	expired leases reclaimed	queue tests

NFR-04: Scalability

Description: The system shall handle growth in projects, citizens, and submissions without redesign.

Technical Analysis:

- Stateless web tier except for the in-process SSE registry; database pooling is configurable.
- Spatial data stored with a PostGIS geometry column for map queries.
- Program scope structure (`constants/agencies.ts`) allows additional programs to be added.

Validation Metrics:

Metric	Target	Measured by
Projects	tens of thousands of rows	analytics limit 30,000
Replicas	supported with the SSE limitation in FR-NOT-01	design review

NFR-05: Maintainability and Quality

Description: The code base shall be easy to change safely.

Technical Analysis:

- TypeScript throughout, Zod validation at boundaries, Drizzle migrations for every schema change.
- About 230 automated test files (Node test runner) cover policy, scope, parsing, analytics, and API behavior.
- ESLint and `tsc --noEmit` are the static checks; documentation lives in `docs/`.

Validation Metrics:

Metric	Target	Measured by
Type check	passes	`tsc --noEmit`
Lint	no errors	`eslint`

NFR-06: Usability and Accessibility

Description: The interfaces shall be understandable and accessible to the public and to staff.

Technical Analysis:

- Meet WCAG 2.1 AA contrast; keyboard focus always visible; interactive targets about 44 by 44 pixels; meaningful text not below readable size.
- Responsive layouts and table affordances for small screens.
- Plain operational wording and explicit unavailable, unknown, stale, and cannot-be-assessed states.
- Guided tours for first-time use; English and Filipino interface.
- ANIA is visually supporting and keeps its verification notice.

Validation Metrics:

Metric	Target	Measured by
Contrast	AA	design review and automated checks
Touch targets	about 44 px	design review

NFR-07: Interoperability

Description: The system shall exchange data in open, documented formats.

Technical Analysis:

- JSON over HTTPS for APIs; CSV export for the directory; GeoJSON and KML for map sources; WMS for GeoServer layers; PSGC codes for locations; S3-compatible object storage.

Validation Metrics:

Metric	Target	Measured by
Export	CSV and JSON	functional test

NFR-08: Auditability and Traceability

Description: Significant actions shall be attributable and reviewable.

Technical Analysis:

- Audit log of changes, security events, blocked uploads, and sync; sync history with counts and errors; moderator and timestamp stored on decisions.
- Requirement IDs trace to code, data, and tests (Chapter 8).

Validation Metrics:

Metric	Target	Measured by
Audit coverage	all staff mutations	code review

NFR-09: Privacy and Compliance

Description: Personal data shall be collected minimally and handled under RA 10173.

Technical Analysis:

- Anonymous E-Reports and public anonymity for feedback; contact details and internal notes never appear in public DTOs.
- Data privacy and data deletion pages explain citizen rights; staff can delete submissions and users.
- Engagement analytics store no visitor identity; chat history expires; demographic survey is optional.
- Analytics cookies are off until the visitor opts in.

Validation Metrics:

Metric	Target	Measured by
Public DTOs	no private fields	unit tests
Retention	chat history purged on schedule	job logs

NFR-10: Localization

Description: The interface shall be usable in English and Filipino.

Technical Analysis:

- Translation sections under `i18n/sections`; style guide for Filipino text; language switch persisted in the browser.

Validation Metrics:

Metric	Target	Measured by
String coverage	both languages	translation tests

NFR-11: Observability and Operations

Description: Operators shall be able to see what the system is doing.

Technical Analysis:

- Sync logs, job progress, audit log, service health page, Docker health check, and optional Google Analytics 4.
- Server logs avoid secrets and personal data.

Validation Metrics:

Metric	Target	Measured by
Health	staff-visible	`/health-check`

NFR-12: Deployability and Portability

Description: The system shall be deployable repeatably with Docker.

Technical Analysis:

- Multi-stage Dockerfile, standalone Next.js output, frozen dependency install, non-root runtime, environment-driven configuration, scripted migrations, nginx sample configuration.

Validation Metrics:

Metric	Target	Measured by
Build	reproducible from `bun.lock`	CI build

4. Data Requirements Analysis

4.1 Purpose

This chapter states what data INFRA Watch must hold, how the data is related, and the integrity, security, and retention rules that apply. The authoritative definitions are the Drizzle schema (`lib/db/schema.ts`, `auth-schema.ts`) and the generated ERD files in `docs/erd/` (`infra-watch.dbml` and `infra-watch.mmd`, editable). The database currently has 44 base tables.

4.2 Data Design Overview

The data design follows six principles:

- **Mirror, do not own, project facts.** `projects` is overwritten by synchronization; local corrections live in separate tables so source values are never lost.
- **Own participation data.** Feedback, comments, votes, issues, responses, SMS review state, and notifications belong to INFRA Watch.
- **Keep identity apart from content.** Authentication tables are Better Auth's; content tables refer to users by identifier.
- **Store flexible payloads as JSON.** Evidence lists, media lists, commodities, notification metadata, and job payloads use `jsonb` so formats can evolve without migrations.
- **Store location once, query it spatially.** Latitude and longitude are kept as plain columns and as a PostGIS `geometry(Point, 4326)` for map queries.
- **Every change is recoverable or attributable.** Moderation fields (who, when, why) and the audit log record decisions; destructive deletes are explicit staff actions.

4.3 Data Entities by Functional Domain

4.3.1 Identity and Access

Entity	Purpose	Requirements
`user`	Account, role, region and program scope, phone, ban state	FR-AUTH-01..06
`session`	Active sessions with IP address and user agent	FR-AUTH-02
`account`	Credential and OAuth provider links (password stored hashed)	FR-AUTH-01, 02
`verification`	One-time codes for email verification and recovery	FR-AUTH-01, 03
`account_preferences`	Per-user notification channel choices	FR-AUTH-04
`user_tour_progress`	Guided tour progress	FR-AUTH-04

4.3.2 Project Mirror, Reference, and Map Data

Entity	Purpose	Requirements
`projects`	Synchronized ABEMIS project (57 columns): identity, location, budget, progress, dates, program, stage, PostGIS point, raw metadata	FR-SYNC-01..03, FR-PUB-01..03
`sync_logs`	One row per synchronization run with counts and errors	FR-SYNC-01, 02
`project_metric_snapshots`	Daily progress and budget snapshot per project	FR-SYNC-04, FR-ADM-01
`project_data_corrections`	Recorded differences between source and corrected values	FR-ADM-04
`project_commodity_overrides`	Staff-approved commodity list for a project	FR-ADM-04
`psgc_locations`	Region, province, municipality, barangay reference (PSGC)	FR-SYNC-05
`regional_statistics`	Yearly regional counts and amounts by program	FR-PUB-04
`map_layers`	GeoServer layer definitions for the map	FR-PUB-03

4.3.3 Citizen Engagement

Entity	Purpose	Requirements
`feedback`	Project feedback with rating, category, media, moderation state, SLA checkpoints	FR-FB-01, 03
`feedback_comments`	Comments on feedback, with auto-rejection data	FR-FB-02
`feedback_comment_votes`	Helpful or unhelpful votes	FR-FB-02
`issues`	E-Reports: ticket, reporter, location, evidence, status, priority, SLA checkpoints	FR-ER-01..03
`issue_responses`	Public or internal staff responses and status changes	FR-ER-03
`sms_grievance_reviews`	Versioned staff review state for each inbound SMS	FR-SMS-01
`contact_messages`, `contact_responses`	Contact and suggestion messages and replies	FR-CON-01
`saved_posts`, `post_subscriptions`	Bookmarks and follow-up subscriptions	FR-FB-02
`submission_surveys`	Optional respondent survey	FR-SUR-01

4.3.4 Notifications

Entity	Purpose	Requirements
`notifications`	One record per event: type, title, message, metadata	FR-NOT-01
`notification_recipients`	Per-user delivery and read state	FR-NOT-01
`notification_reads`	Read receipts	FR-NOT-01

4.3.5 Content and Configuration

Entity	Purpose	Requirements
`articles`	Linked news and updates	FR-PUB-05
`announcements`	Dated announcement banners	FR-PUB-05
`live_videos`	Facebook, YouTube, or recorded broadcast entries	FR-PUB-05
`email_templates`	Editable email subject and body with variables	FR-ADM-07
`system_settings`	Key-value settings	FR-ADM-07

4.3.6 AI and Knowledge

Entity	Purpose	Requirements
`ai_chat_history`	Chat turns with provider, model, tool names, token counts, status	FR-AI-01
`ai_chat_rate_limits`	Rolling-window request counters	FR-AI-01
`kb_documents`	Knowledge documents and FAQ entries, with processing status	FR-AI-02
`kb_chunks`	Text chunks with 384-dimension embedding stored as JSON	FR-AI-02
`mydas_dashboards`	Saved MYDAS dashboards	FR-AI-03

4.3.7 Operations and Platform

Entity	Purpose	Requirements
`audit_logs`	Who changed what, with old and new values	FR-ADM-06
`background_jobs`	Queue with leases, attempts, progress, result	FR-SYNC-01, FR-NOT-01
`upload_sessions`	Resumable upload sessions	FR-UP-01
`analytics_events`, `analytics_daily_aggregates`	Privacy-minimal usage events and daily rollups	FR-ADM-09
`spatial_ref_sys`	PostGIS coordinate reference table (installed by the extension)	Platform

4.4 Entity Relationships Summary

Sixteen foreign keys are enforced in the database. Most of them point at `projects.abemis_id`, the stable source identifier, rather than the internal UUID. References from content to users (`user_id`, `reporter_user_id`) are logical and not enforced, which keeps feedback and reports intact if an account is removed. Figure 3 shows the engagement and project entities; Figure 4 shows identity, notification, knowledge, and contact entities.

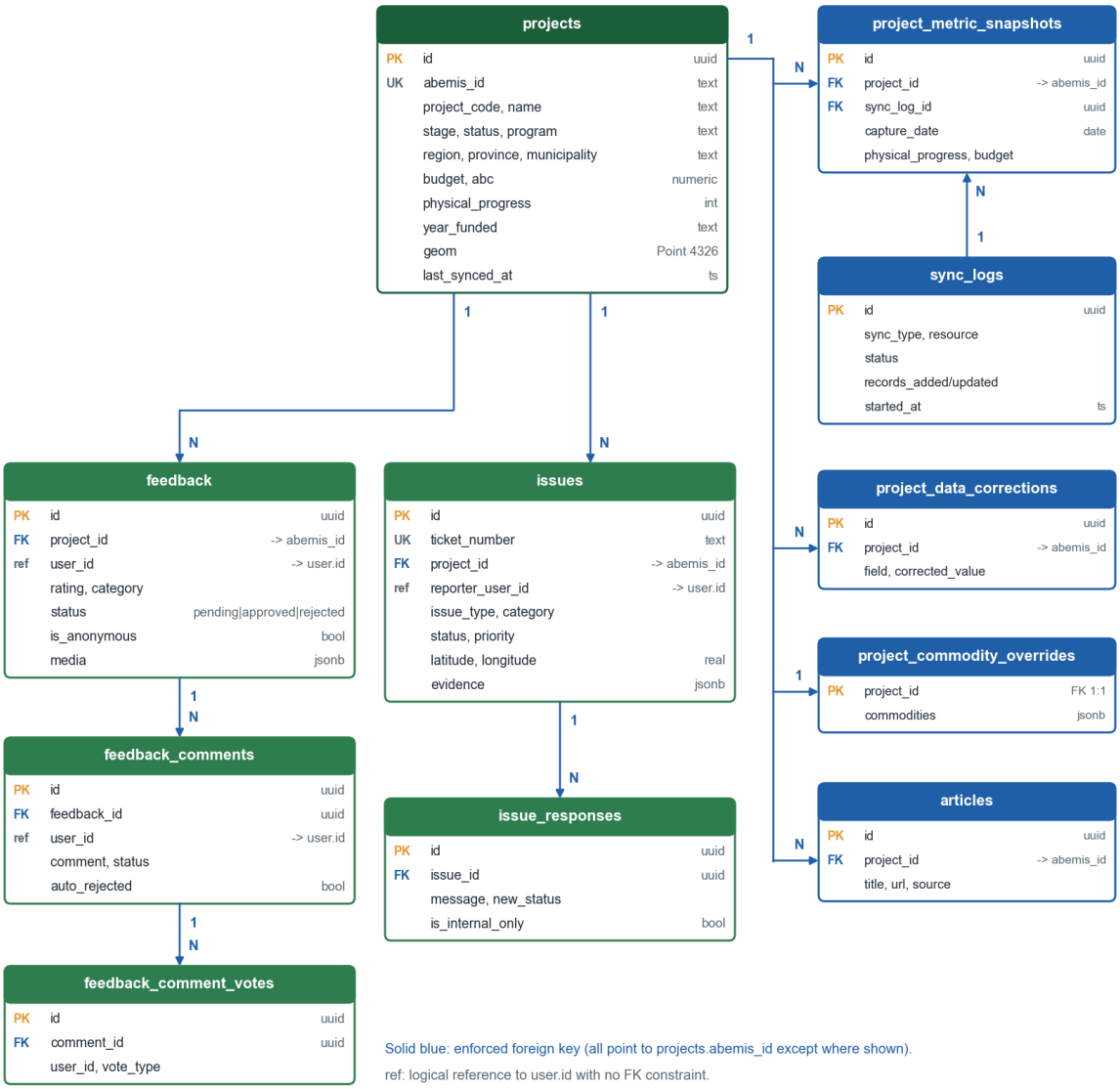


Figure 3. Core entities: projects, citizen engagement, and synchronization.

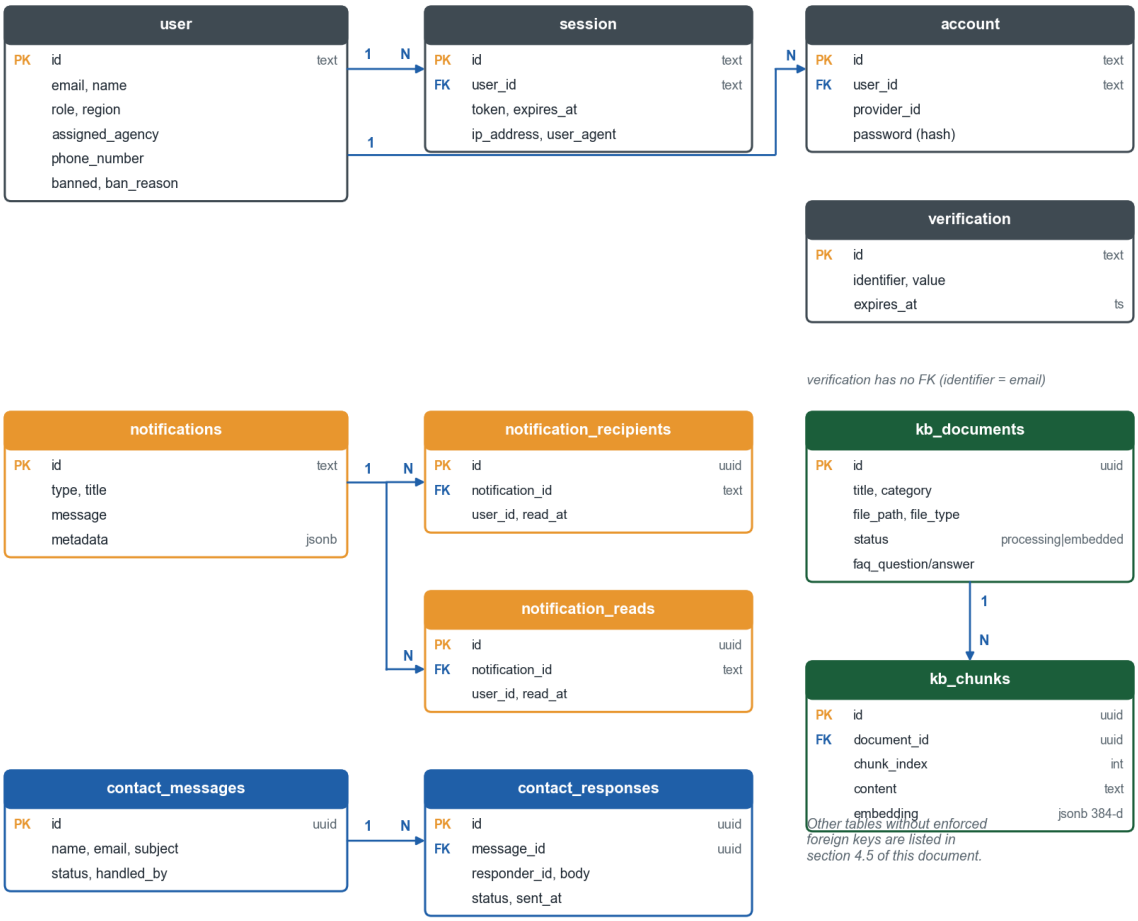


Figure 4. Supporting entities: identity, notifications, knowledge base, and contact.

4.5 Data Integrity and Constraints

Rule	Mechanism
A project is unique by `abemis_id` and by `abemis_raw_id`	Unique constraints on `projects`
An issue has a unique ticket number	Unique constraint on `issues.ticket_number`
Effect of removing a project	`issues.project_id` and `articles.project_id` use `ON DELETE SET NULL`; `feedback.project_id` uses `CASCADE`
Deleting a feedback removes its comments and votes	`CASCADE` from `feedback_comments` and `feedback_comment_votes`
Deleting an issue removes its responses	`CASCADE` from `issue_responses`
Corrections to a project cannot be orphaned	`project_data_corrections.project_id` uses `RESTRICT`
Deleting a user removes sessions and accounts	`CASCADE` from `session` and `account`
Knowledge chunks follow their document	`CASCADE` from `kb_chunks.document_id`; unique (`document_id`, `chunk_index`)

Rule	Mechanism
Only one active background job per dedupe key	Partial unique index `background_jobs_active_dedupe_uidx`
Status fields hold known values	Application-level validation (Zod) and status constants; database columns are free text

Housekeeping The database currently carries two identical foreign keys on `kb_chunks.document_id` (`kb_chunks_document_id_fkey` and `kb_chunks_document_id_kb_documents_id_fk`), and two equivalent unique indexes on (`document_id`, `chunk_index`). One of each can be dropped in a cleanup migration.

4.6 Data Security Considerations

- Passwords are stored as hashes in `account`; they never appear in logs, API responses, or the audit log.
- Personal data in `user`, `issues` (reporter name, contact, email), `feedback` (user identifier), and `contact_messages` is exposed only to authorized staff within scope; public data transfer objects remove it.
- Evidence files are stored outside the database; paths are saved in `issues.evidence` and `feedback.media`. Issue evidence and knowledge-base files sit in a private bucket.
- `audit_logs` stores IP address and user agent for security events and is visible only to authorized staff.
- Secrets (database URL, auth secret, SMS and SMTP credentials, API keys) are provided only through environment variables.

4.7 Data Retention and Archiving

Data	Retention rule
`ai_chat_history`	Kept for the configured period (`CHAT_HISTORY_RETENTION_DAYS`, maximum 365 days); expired rows are purged by a daily 03:15 job
`project_metric_snapshots`	Pruned after 365 days
`analytics_events`	Pruned by the engagement analytics maintenance job; daily aggregates retained
`upload_sessions`	Expire and are cleaned up by the background worker
`verification` codes	Expire after `EMAIL_OTP_EXPIRY` seconds
`feedback`, `issues`, responses	Kept until staff or the owner deletes them; deletion is permanent and audited
`audit_logs`	Kept indefinitely unless an administrator deletes entries; no automatic purge
`sync_logs`	Kept; used as sync history

4.8 Data Validation Rules

Field	Rule
Email	Well-formed, lower-cased, at most 254 characters
Feedback comment	Required; the server accepts up to 1,000 characters (the composer allows 2,000, see FR-FB-01)
Feedback rating	Optional integer 1 to 5
Feedback category	One of general, quality, progress, concerns
Feedback and comment media	Paths must match the portal's own pattern (`feedback/<timestamp>-<hex>.<ext>`)
Comment text	1 to 10,000 characters
Suggestion subject, message, name	160, 5,000, and 120 characters maximum
Coordinates	Valid latitude and longitude inside Philippine bounds to be plotted
Upload type and size	Allowlisted MIME types and extensions with matching file signature; image and video size limits from environment
Live video URL	HTTPS provider URL; schedule and state must be compatible
Directory page size	Clamped to 10 to 100 on admin lists; 20 on the public directory
Chat request	At most 64 KB, 20 history messages, 4,000 characters per message

5. Interface Requirements Analysis

5.1 Purpose

This chapter defines the interfaces INFRA Watch must provide to people and consume from other systems. Detailed endpoint contracts are in the SDD (Chapter 5).

5.2 User Interface Requirements

5.2.1 Page Inventory

Area	Routes	Access
Public portal	`/`, `/projects`, `/projects/[id]`, `/map`, `/evidence-map`, `/infra-analytics`, `/statistics`, `/citizen-feed`, `/articles-and-updates`, `/live`	Everyone
Public reporting	`/report-issue`, `/report-issue/new`, `/report-issue/[id]`, `/report-issue/sms`, `/citizen-guides/sms`, `/suggestions`, `/contact`	Everyone (submission of feedback requires sign-in)
Information	`/about`, `/faq`, `/data-privacy`, `/data-deletion`, `/terms-of-service`	Everyone
Authentication	`/sign-in`, `/sign-up`, `/forget-password`	Everyone
Citizen account	`/my-profile`, `/account-settings`, `/my-feedbacks`, `/my-issues`, `/my-issues/[id]`, `/my-notifications`	Signed-in users
Staff console	`/dashboard`, `/admin-projects`, `/feedbacks`, `/comment-moderation`, `/issues`, `/issues/sms-review`, `/contact-messages`, `/data-quality`, `/audit-logs`, `/sync`, `/reports/*`, `/knowledge-base`, `/articles`, `/announcements`, `/live-videos`, `/map-layers`, `/email-templates`, `/regional-statistics`, `/settings`, `/health-check`	Moderator, regional admin, admin (some admin-only)
Admin-only tools	`/user-management`, `/mydas`, `/executive-brief`, SLA reports, sync trigger	Admin

5.2.2 Interaction and Design Requirements

- Directory and map filters are held in the URL so a view can be shared and restored.
- Forms validate on the client and again on the server with the same rules; errors are shown next to the field.
- Long operations show progress (sync, upload, knowledge-base indexing). Server-sent events deliver notifications and job progress.
- Tables degrade to readable layouts on narrow screens; charts always have a text alternative or table.
- Every data state has an explicit visual: ready, empty, unavailable, unknown, stale, or cannot be assessed.
- ANIA appears as a supporting assistant with its verification notice; it never replaces the primary navigation.
- No ornamental gradients, generic sparkle imagery, excessive rounded cards, or tiny meaningful text (see `docs/06-ui-ux-design.md`).

5.3 External System Interfaces

5.3.1 ABEMIS

Item	Specification
Endpoint	`${ABEMIS_BASE_URL}${ABEMIS_INFRA_ENDPOINT}`, default path `/api/infra-amefip-list`
Authentication	`x-api-key` header from `ABEMIS_API_KEY`
Request	`GET` with `page`, `page_size` (500 during sync), optional `year_funded`
Response	JSON list with a `pagination.total_pages` value; each record carries identifiers, name, location, budget, progress, dates, stage, status, program, contractor, and media references
Behavior	Up to three retries with back-off; redirects refused; the target must stay on the configured origin
Frequency	Every three hours, skipped if a sync succeeded in the last 20 hours; manual on demand

5.3.2 SMS Gateway and SMS Grievance Feed

Item	Specification
Outbound SMS	`POST` JSON to `SMS_API_URL` with sender ID, username, password, normalised mobile number (`+639XXXXXXXXX`), and message; success is judged by the HTTP status
Inbound feed	`GET` JSON of untriaged messages from `SMS_GRIEVANCE_API_URL`: identifier, SMS identifier, sender number, content, receive date and time, location, status, month, year
Fallback	If the feed is unreachable, labelled synthetic records are shown so staff can still use the screen
Privacy	Sender numbers are staff-only; sample data contains no real numbers

5.3.3 Email, Maps, Storage, Identity, and AI

System	Interface
SMTP	`MAIL_HOST`, `MAIL_PORT`, credentials, sender name and address; STARTTLS required on port 587
GeoServer	`NEXT_PUBLIC_GEOSERVER_URL` and workspace; WMS layers listed in `map_layers`; the map hides overlays if the URL is empty
MinIO	S3-compatible API with separate public and private buckets; start-up reconciles bucket policies and refuses to run if the private bucket allows anonymous access
Google OAuth	Authorization-code flow through Better Auth when client credentials are set
AI providers	`AI_PROVIDER` selects Gemini (default), OpenAI, Anthropic, or Kimi; `AI_MODEL` overrides the model
Embeddings	Local sentence-embedding model (all-MiniLM-L6-v2), 384 dimensions, run inside the application
Google Analytics 4	Measurement ID inlined at build time; loaded only after analytics consent
Redis	`REDIS_URL`; connection failures are tolerated and retried after 30 seconds

5.4 Hardware Interfaces

INFRA Watch has no direct hardware interface. Citizens use their own phone or computer; the browser may use the device camera and location services when the citizen chooses to attach a photo or share coordinates. The voice assistant uses the administrator's microphone through the browser.

5.5 Software Interfaces

Component	Role
PostgreSQL 18 + PostGIS	System of record; `postgres` driver; Drizzle ORM and drizzle-kit migrations
Better Auth	Authentication, sessions, admin and access-control plugins
Next.js 16 / React 19	Rendering, routing, server actions, route handlers
MinIO client	Object storage access
Nodemailer	SMTP delivery
Vercel AI SDK	Model access for Gemini, OpenAI, Anthropic, and Kimi
Leaflet, Recharts	Map and chart rendering in the browser
node-cron	Schedules inside the application process
TensorFlow.js (NSFWJS), sharp, ffmpeg	Image and video checks, image handling

5.6 Communication Protocols and Standards

- HTTPS with TLS for all public traffic; nginx redirects HTTP to HTTPS.
- JSON request and response bodies; multipart for uploads; server-sent events for notification and job streams.
- Standard geospatial formats: WGS 84 (SRID 4326), GeoJSON, KML, WMS.
- Cookie-based sessions with `SameSite` and `Secure` settings in production; Cross-Origin-Opener-Policy `same-origin` and Cross-Origin-Embedder-Policy `credentialless` headers.

5.7 Interface Security and Compliance

- Authentication and role checks on every staff interface; mutating staff endpoints reject cross-origin requests.
- JSON bodies are size-limited (32 KB default) and must use `application/json`.
- Credentials for external systems stay on the server and are never sent to the browser.
- Public APIs return only public-safe fields; private evidence is served through authenticated handlers.
- Per-client and global limits protect the chat and analytics endpoints; nginx replaces client-supplied forwarding headers so rate-limit identity cannot be spoofed.

5.8 Interface Traceability

Interface	Requirements
ABEMIS API	FR-SYNC-01, 02, 03, 04
SMS gateway and feed	FR-SMS-01, FR-NOT-01, FR-NOT-02

Interface	Requirements
SMTP	FR-AUTH-01, 03, FR-CON-01, FR-NOT-01
GeoServer and KML	FR-PUB-03
MinIO	FR-UP-01, FR-ER-01, FR-FB-01, FR-AI-02, FR-PUB-05
AI providers and embeddings	FR-AI-01 to 04, FR-ADM-02
Google OAuth	FR-AUTH-02
Redis and in-process cache	NFR-01, NFR-03

6. System Constraints and Assumptions

6.1 Purpose

This chapter lists the limits the design must live within and the assumptions that were made. If an assumption stops holding, the related requirements and risks in Chapter 7 should be revisited.

6.2 Technical Constraints

Area	Constraint
Hosting	Single Docker service bound to loopback behind nginx; database, object storage, and reverse proxy are external services
Stack	TypeScript, Next.js App Router, React 19, Better Auth, Drizzle ORM, Bun runtime
Database	PostgreSQL with PostGIS; schema changes only through Drizzle migrations; `pgvector` is not used (embeddings are stored as JSON and compared in the application)
Security	Secrets only in environment variables; production refuses to start without auth secret and URL
Performance	Aggregations capped at 30,000 rows; list endpoints paginated
Integration	ABEMIS is read-only; one endpoint; one API key
Scheduling	Schedules run in the application process; jobs execute through the database queue

6.3 Environmental Constraints

- **Hardware:** a Linux host able to run Docker, PostgreSQL, and MinIO; ffmpeg is bundled in the application image.
- **Software:** Docker with an external network shared with the other services; nginx with Let's Encrypt certificates.
- **Network:** outbound HTTPS to ABEMIS, the SMS gateway, the SMTP server, AI providers, and (for the browser) GeoServer.
- **Time:** schedules assume the Asia/Manila time zone.

6.4 Operational Constraints

- Staff must assign region and program to each moderator; otherwise the moderator sees no project data.
- Voice, managerial AI, and the AI assistant require environment settings and, for voice, extra services.
- Database and object-storage backups are an operational responsibility outside the application.
- Moderation capacity must match the 72-hour standard; the system reminds but cannot staff.

6.5 Assumptions

Group	Assumption
Infrastructure	PostgreSQL, MinIO, and nginx are provisioned, monitored, and backed up by BAFE or its host.
Users and organization	BAFE appoints administrators and moderators and maintains their assignments; citizens can use a modern browser or SMS.
Data and workflow	ABEMIS stays the system of record; INFRA Watch is refreshed rather than edited; data-quality findings are fixed in ABEMIS.
Integration	ABEMIS, SMS gateway, SMTP, and AI providers keep their current interfaces and credentials remain valid.
Security	TLS is terminated at nginx; the application is reachable from the internet only through it.

6.6 Dependencies

Dependency	Impact if unavailable
ABEMIS API	Public data stops refreshing; pages continue to show the last data with its sync time
PostgreSQL	System unavailable
MinIO	Uploads and media views fail (or fall back to local storage); text content continues
SMTP server	No verification, recovery, or email notification; new accounts cannot be activated
SMS gateway	No SMS replies or alerts; staff review of the feed still works
AI provider	Assistant, executive brief, and MYDAS unavailable; the rest of the portal is unaffected
GeoServer	Map overlays hidden; project markers still shown

6.7 Risks Related to Constraints

The constraints above give rise to the operational and integration risks R-01, R-04, R-08, R-14, R-17, and R-18 in Chapter 7.

7. Risk Analysis and Mitigation Plan

7.1 Purpose

This chapter records the main risks to the system and the controls that reduce them. Items marked *review* were identified in the 27 September 2026 code, security, and workflow reviews; their current status should be confirmed against the repository before release decisions.

7.2 Risk Assessment Methodology

Each risk is scored for likelihood (L) and impact (I) on a three-point scale (1 low, 2 medium, 3 high). The rating is L multiplied by I: 1 to 2 Low, 3 to 4 Medium, 6 to 9 High.

7.3 Risk Categories

Technical (T), Operational (O), Integration (I), Security and Compliance (S), and Organizational (H) risks are listed together in the register below.

7.4 Risk Register

ID	Cat.	Risk	L	I	Rating	Mitigation
R-01	I	ABEMIS is unavailable or returns errors (recurring upstream outages have been observed).	3	2	High	Three-hourly schedule with 20-hour skip rule; stale-run recovery; data pages show the last sync time; sync history for follow-up with ABEMIS owners.
R-02	I	ABEMIS changes stage or status wording and public statuses become wrong.	2	2	Medium	Mapping table plus keyword fallback; data-quality module flags unclassified records; transform tests.
R-03	T	A source-ID correction path can delete and recreate a project when foreign keys block renaming, cascading to dependent feedback and snapshots (*review*).	1	3	Medium	Replace delete-and-recreate with a key-update migration; add a guard test; confirm remediation before the next release.
R-04	O	When MinIO is unreachable the upload path can fall back to local storage that is lost on container restart.	2	3	High	Fail uploads visibly instead of falling back in production; monitor MinIO; include storage in backups.
R-05	S	Stored photos keep EXIF metadata, including GPS, so a public image may reveal more location detail than intended.	2	2	Medium	Strip metadata on upload for public media (keep a server-side copy of coordinates); state the behavior in the privacy notice.
R-06	S	Knowledge documents have no draft or published state; any active embedded document can be quoted by the public assistant.	2	3	High	Add a visibility field and require publication before retrieval; staff guidance not to upload confidential files meanwhile.
R-07	T	Issue status can move between any recognised states, and response, status, audit, and notification writes are not one transaction.	2	2	Medium	Define an allowed-transition map; wrap the write set in a transaction; test partial failure.
R-08	T	Live notifications use an in-process registry, so extra replicas would miss events for users connected elsewhere.	1	2	Low	Stay single-replica, or move fan-out to Redis pub/sub or Postgres LISTEN/NOTIFY before scaling out.

ID	Cat.	Risk	L	I	Rating	Mitigation
R-09	T	SLA reminder checkpoints can be stamped before delivery succeeds, preventing automatic retry.	2	2	Medium	Stamp after confirmed delivery or record delivery status separately.
R-10	S	AI answers may be wrong, expose internal details, or run up cost.	2	2	Medium	Bounded read-only tools, refusal policy, rate limits, verification notice, history retention limits; monitor token usage.
R-11	S	Regional admins could bypass region limits through direct administrative auth endpoints (*review*).	1	3	Medium	Restrict direct plugin administration and keep scoped server actions; confirm the patch is applied.
R-12	S	Runtime dependencies with published advisories (*review*).	2	3	High	Update pinned versions, run an audit in CI, and review overrides regularly.
R-13	S	Citizens may assume "anonymous" feedback is invisible to staff, while the account identifier is retained for accountability.	2	2	Medium	State the rule on the form, in the privacy notice, and in the terms of service.
R-14	H	A moderator without region or program assignment sees nothing, which can look like an outage.	2	1	Low	Show a clear unassigned-scope message; include assignment in onboarding checklist.
R-15	H	Staff capacity is insufficient for the 72-hour response standard.	2	3	High	Reminders at 24 and 60 hours, SLA reports, admin escalation; review staffing against report data.
R-16	I	SMS gateway outage, cost, or credential leak.	1	2	Low	Decisions saved before sending; failed sends visible; credentials in environment only; rotate on suspicion.
R-17	O	Voice features are enabled without the required wake-word and Whisper services.	1	1	Low	Keep `VOICE_ASSISTANT_ENABLED` off until services exist; document the sidecars.
R-18	O	Single-host deployment: loss or corruption of database or object storage.	1	3	Medium	Scheduled database and bucket backups with restore tests; migration scripts are versioned; rollback SQL kept in `docs/migrations`.
R-19	S	With auto-accept on (the default), feedback and issue descriptions are published after automated checks only; harmful text or media that passes the checks is visible until a moderator acts.	2	2	Medium	Keep text and media checks fail-closed; moderators review the newest items daily; auto-accept can be switched off in `/settings` during sensitive periods.
R-20	T	Limits disagree between layers (feedback composer allows 2,000 characters; the server rejects over 1,000), so citizens can lose a long comment on submit.	2	1	Low	Align the composer limit with the server limit and show the remaining count.

7.5 Risk Monitoring and Governance

- Review the register at each release and after any security review; record status changes in the change history.
- Track operational risks (R-01, R-04, R-15, R-18) with the sync history, health check, SLA reports, and backup logs.
- Assign an owner to every High risk and confirm the mitigation before announcing a major release.

7.6 Residual Risk and Contingency

After mitigation the main residual risks are dependence on ABEMIS availability and on staff capacity to respond within 72 hours. Contingency: the portal continues to serve the last synchronized data with a visible sync time; unanswered items are surfaced to administrators by reminders and SLA reports; SMS review gives staff a path for citizens without internet access.

8. Requirements Traceability Matrix (RTM)

8.1 Purpose

The RTM links each requirement to where it is analysed, the data it uses, the interface that exposes it, and how it is accepted. It supports impact analysis when a requirement or table changes.

8.2 Traceability Methodology

Field	Description
Requirement ID	Identifier defined in Chapter 3
RAD section	Where the requirement is analysed
Data entities	Tables read or written
Interfaces	Routes, pages, or jobs
Acceptance criteria	Observable condition confirming the requirement

8.3 Functional Requirements Traceability

Req. ID	Sec.	Data entities	Interfaces	Acceptance criteria
FR-AUTH-01	3.1	user, account, verification, session	/sign-up, /api/auth	Unverified account cannot sign in; code expires after the configured time.
FR-AUTH-02	3.1	session, account, audit_logs	/sign-in, /api/auth	Valid credentials create a session; banned user is refused.
FR-AUTH-03	3.1	verification, account	/forget-password	Valid code allows a new password; expired code is rejected.
FR-AUTH-04	3.1	user, account_preferences	/my-profile, /account-settings	Only the owner changes phone and preferences.
FR-AUTH-05	3.1	user, projects	/api/admin/*, (admin)	Citizen gets 403; scoped moderator cannot open out-of-scope records.
FR-AUTH-06	3.1	user, session, audit_logs	/user-management	Self-ban, self-delete, self-role-change are refused; each action audited.
FR-SYNC-01	3.2	projects, sync_logs, background_jobs	scheduler, ABEMIS API	Run logged; skipped within 20 h of success; stale run recovered.
FR-SYNC-02	3.2	sync_logs, background_jobs	/sync, POST /api/admin/sync	Only admin triggers; progress streams; history visible.
FR-SYNC-03	3.2	projects	year-scope, stage mapping	Out-of-scope years, cancelled and invalid records never appear.
FR-SYNC-04	3.2	project_metric_snapshots	/dashboard	Snapshot after successful sync only; older than 365 days pruned.
FR-SYNC-05	3.2	psgc_locations	/locations/sync	Location filters populate after the job.
FR-PUB-01	3.3	projects, sync_logs	/projects, /api/projects	Search, filter, sort, page 20; last-sync time shown.
FR-PUB-02	3.3	projects, feedback	/projects/[id]	Resolves by id, code, or UUID; missing values shown as unavailable.
FR-PUB-03	3.3	projects, map_layers	/map, /api/map-layers	Only valid Philippine coordinates plotted; overlays toggle.

Req. ID	Sec.	Data entities	Interfaces	Acceptance criteria
FR-PUB-04	3.3	projects, regional_statistics	/infra-analytics, /statistics	Totals match directory; >30,000 rows returns unavailable.
FR-PUB-05	3.3	articles, announcements, live_videos	/articles-and-updates, /live	Inactive or expired entries hidden; one live and one featured max.
FR-PUB-06	3.3	none	/about, /faq, /data-privacy, ...	Pages exist in both languages; consent banner works.
FR-PUB-07	3.3	projects	/api/public/project-export	CSV and JSON download with the same filters.
FR-FB-01	3.4	feedback, notifications	/citizen-feed, POST /api/projects/[id]/feedback	Stored as approved (auto-accept on) or pending (off); citizen acknowledged; staff notified.
FR-FB-02	3.4	feedback_comments, feedback_comment_votes	server actions	One vote per user per comment; flagged comment auto-rejected.
FR-FB-03	3.4	feedback, audit_logs	/feedbacks, /api/admin/feedback	Approve or reject stores moderator, time, note; audit written.
FR-ER-01	3.4	issues	/report-issue/new, POST /api/issues	Ticket number returned; status reviewing (auto-accept on) or pending; staff notified.
FR-ER-02	3.4	issues, issue_responses	/report-issue, /my-issues	Public view lacks contact and internal notes.
FR-ER-03	3.4	issues, issue_responses, audit_logs	/issues, /api/admin/issues	Response saved; resolved sets resolved_at; reporter notified.
FR-UP-01	3.4	upload_sessions, audit_logs	/api/upload	Wrong type or signature rejected and audited; evidence private.
FR-GEO-01	3.4	feedback, issues	/evidence-map, /api/evidence	Only approved or public-safe records appear.
FR-SMS-01	3.4	sms_grievance_reviews	/issues/sms-review	Decision saved before SMS; failed send shown; version conflict blocks duplicate reply.
FR-CON-01	3.4	contact_messages, contact_responses	/contact, /contact-messages	Message stored; staff reply recorded.
FR-SUR-01	3.4	submission_surveys	/api/submission-survey	Survey can be skipped without blocking submission.
FR-NOT-01	3.4	notifications, notification_recipients	/my-notifications, SSE	Recipient sees item live and in history; read state persists.
FR-NOT-02	3.4	issues, feedback	sla-followup job	Reminders at 24 and 60 hours; breach notice after 72.
FR-ADM-01	3.5	projects, project_metric_snapshots	/dashboard	Moderator sees only scope; invalid filters rejected.
FR-ADM-02	3.5	ai_chat_history	/executive-brief	Available only when enabled; shows verification notice.
FR-ADM-03	3.5	projects	/admin-projects	Scope enforced.
FR-ADM-04	3.5	projects, project_data_corrections	/data-quality	Findings exportable; PATCH returns 405.
FR-ADM-05	3.5	issues, feedback	/reports/issues, /reports/feedbacks	Admin only; auto-acknowledgment excluded.
FR-ADM-06	3.5	audit_logs	/audit-logs	Failed logins and blocked uploads listed.
FR-ADM-07	3.5	articles, announcements, email_templates, map_layers, live_videos, system_settings	(admin) content pages, /settings	Changes audited; HTTPS-only live URLs; maintenance mode lets staff in.
FR-ADM-08	3.5	audit_logs	/health-check, /api/health	Staff-only; cache clear audited.
FR-ADM-09	3.5	analytics_events, analytics_daily_aggregates	/reports/citizen-engagement	Aggregates only; no visitor identity.
FR-AI-01	3.6	ai_chat_history, ai_chat_rate_limits	POST /api/chat	Limits enforced; out-of-scope request refused; answers use tools.
FR-AI-02	3.6	kb_documents, kb_chunks	/knowledge-base	Archived documents not retrieved; reindex keeps old chunks until swap.

Req. ID	Sec.	Data entities	Interfaces	Acceptance criteria
FR-AI-03	3.6	mydas_dashboards	/mydas	Non-SELECT or off-allowlist SQL rejected.
FR-AI-04	3.6	none	/api/voice/*	Admin only; works only when enabled.

8.4 Non-Functional Requirements Traceability

NFR	Primary satisfied by	Verified by
NFR-01 Performance	Pagination, row limits, cache, indexes	Query tests; performance checks in browser
NFR-02 Security	Permissions, scope, upload validation, AI guards	Scope and policy tests; security review
NFR-03 Reliability	Job queue leases, schedule skip logic, restart policy	Queue tests; sync logs
NFR-04 Scalability	Stateless web tier, pooling, PostGIS	Design review
NFR-05 Maintainability	TypeScript, Zod, migrations, tests	`tsc`, `eslint`, test suite
NFR-06 Usability and accessibility	Design contract, responsive layout, explicit data states	Design review; contrast and focus checks
NFR-07 Interoperability	JSON, CSV, GeoJSON, KML, WMS, PSGC	Functional tests
NFR-08 Auditability	Audit log, moderation fields, sync logs	Code review
NFR-09 Privacy	Public DTOs, anonymity, retention, consent	Unit tests; privacy review
NFR-10 Localization	i18n sections, language toggle	Translation tests
NFR-11 Observability	Health check, logs, GA4	Operational review
NFR-12 Deployability	Dockerfile, compose, start script, nginx sample	Build and deploy rehearsal

8.5 Module, Entity, and Interface Summary

Module	Main tables	Main interfaces
Authentication and access	user, session, account, verification	/api/auth, guards
Project sync	projects, sync_logs, project_metric_snapshots, background_jobs	/api/admin/sync, jobs
Public portal	projects, regional_statistics, map_layers, articles	/api/projects, /api/public/*
Citizen engagement	feedback*, issues, issue_responses, sms_grievance_reviews	/api/issues, /api/projects/[id]/feedback, /api/upload
Notifications	notifications, notification_recipients	/api/notifications
Staff console	audit_logs, sync_logs, system_settings	/api/admin/*
AI services	ai_chat_history, kb_documents, kb_chunks, mydas_dashboards	/api/chat, /api/knowledge-base/*

9. Appendices

9.1 Glossary of Terms

Term	Definition
ABC	Approved Budget for the Contract, as recorded in ABEMIS
ABEMIS	Agricultural and Biosystems Engineering Management Information System; the source of project data
AMEFIP	Agricultural Machinery, Equipment, Facilities, and Infrastructures Program
ANIA	The INFRA Watch AI assistant
Audit log	Record of who changed what and when
BAFE	Bureau of Agricultural and Fisheries Engineering
E-Report	Citizen report of an infrastructure issue, tracked by ticket number
Feedback	Citizen opinion or observation about a specific project, public once approved
INS	Irrigation Network Services, a sub-program of AMEFIP
Moderator scope	Region and program assignment that limits what a moderator can see and act on
MYDAS	Natural-language dashboard designer for administrators
Public status	Simplified project status: Not yet started, On going, Completed
Read model	A copy of source data kept for fast, safe reading
SLA	Service-level agreement; here, the 72-hour response standard
Snapshot	Daily record of a project's progress and budget

9.2 Acronyms

Acronym	Meaning	Acronym	Meaning
API	Application Programming Interface	PSGC	Philippine Standard Geographic Code
DTO	Data Transfer Object	RBAC	Role-Based Access Control
EXIF	Exchangeable Image File Format	SMS	Short Message Service
GA4	Google Analytics 4	SMTP	Simple Mail Transfer Protocol
GPS	Global Positioning System	SSE	Server-Sent Events
JSON	JavaScript Object Notation	TLS	Transport Layer Security
KML	Keyhole Markup Language	WCAG	Web Content Accessibility Guidelines
OTP	One-Time Password	WMS	Web Map Service
RA	Republic Act	NSFW	Not safe for work (content check)

9.3 Change History

Version	Date	Author	Description
1.0	October 2026	INFRA Watch Project Team	First formal RAD, written from the implemented system and verified against the repository.

9.4 Document Control

Item	Detail
Document	Requirements Analysis Document (RAD), INFRA Watch
Version	1.0
Date	October 2026
Prepared by	INFRA Watch Project Team
Institution	Department of Agriculture - Bureau of Agricultural and Fisheries Engineering (DA-BAFE)
Companion document	Software Design Document (SDD), INFRA Watch
Review and approval	Reviewed by: _____ Approved by: _____ Date: _____